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DEPARTMENT OF COMPUTING

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# MultiRAG: A Framework to Fuse Expert Retrievers via Hybrid Reranking

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Submitted in partial fulfillment of the requirements for the MSc degree in  
Computing of Imperial College London

September 2025

## Abstract

Retrieval-Augmented Generation (RAG) is a key technique for grounding Large Language Models (LLMs) in factual data. However, the performance of RAG systems is often constrained by the monolithic nature of their underlying retriever, as no single method excels across all data types and query complexities. Real-world applications often involve a heterogeneous mix of data modalities (e.g. text, tables, images), which poses a significant challenge for unimodal retrieval systems. This diversity necessitates a hybrid approach that can effectively fuse the outputs of multiple, specialised retrievers to provide a more comprehensive context for the generator. Furthermore, evaluating such systems is hampered by a lack of robust, domain-specific benchmarks.

To address this, we propose MultiRAG, a new architectural paradigm that aggregates and reranks candidate documents from multiple, independent retrieval systems, spanning both textual and multimodal sources before the final generation step. We evaluate MultiRAG’s performance on 10 established benchmarks including ViDoRe datasets, BEIR datasets, and a new, robust, domain-specific dataset. We demonstrate its superiority over ColPali, a state-of-the-art multimodal retrieval baseline.

To facilitate more rigorous evaluation in specialised fields, we also introduce an LLM-driven methodology for creating domain-specific RAG benchmarks. Using a ‘Consulting’ dataset of  $\sim 16,000$  pages generated with this method, our extensive ablation studies show that MultiRAG improves retrieval metrics and reduces the performance bias observed when single retrievers are applied to non-optimal document types.

Overall, our findings suggest that a hybrid, reranking-based architecture is an effective and robust strategy for optimising RAG systems in complex information environments. Furthermore, our dataset generation pipeline offers a scalable tool for future research in domain-specific RAG.

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## Acknowledgments

I would like to express my most sincere gratitude to my supervisor, Dr Nihir Vedd, for his invaluable guidance and support throughout this research project. His deep expertise in natural language processing and retrieval systems, and his insightful feedback, were instrumental in shaping the direction of this work. His rigorous approach and challenging questions pushed me to refine my ideas and improve the quality of my research. This project would not have been possible without his supervision.

I would also like to thank my second marker, Dr Chiraag Lala, for his constructive feedback and encouragement. His perspectives helped emphasise the importance of a rigorous evaluation framework for RAG systems, as well as the need for motivating real-world applications.

Finally, I am grateful to my peers at Imperial College who shared their own challenges and enthusiasm to remain motivated during this journey.

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# Chapter 1 – Introduction

## 1.1 The Foundational Models Revolution

In 2017, the introduction of the transformer architecture[60] at the NeurIPS conference sparked widespread academic and industry interest in large language models (LLMs). The transformer mechanism fundamentally reshaped sequence modeling. This breakthrough drove a series of state-of-the-art models the following year. Among them were BERT[29] presented by Google, which excelled at language understanding for classification tasks, and GPT[21] proposed by OpenAI which showcased fluent language generation.

Intensified research effort led to a computer vision breakthrough as well a couple of years later. OpenAI introduced a novel approach to learning visual representation that scaled effectively across tasks named CLIP[50], which popularised vision language models (VLMs). Unlike earlier pioneering research DeViSE[33] which lacked contextual language modeling, or ViLBERT[42] which required task-specific fine-tuning preventing it from generalising well, training on large-scale image-text pairs using a contrastive objective enabled CLIP to remain robust and performant across a wide range of visual tasks.

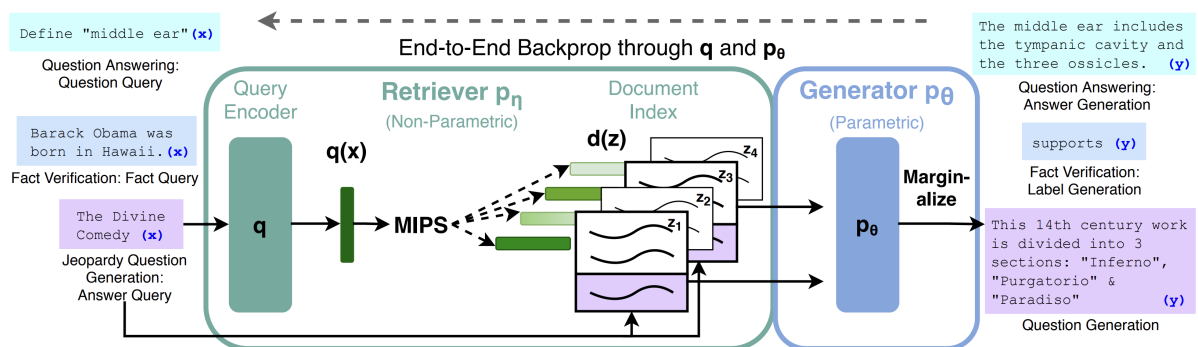
The surge of foundational models[20] gained further public attention in 2022 with the release of ChatGPT[6], Stable Diffusion[13] and DALL-E[2]. These generative models and their optimised variants are now widely deployed across industry. Their remarkable capacity to generate coherent text and high-fidelity images, as well as perform reasoning and handle knowledge-intensive tasks, has transformed professional workflows[4]. A 2025 McKinsey Global Survey on AI shows the adoption of generative AI in business has surged dramatically, with 71% of organizations reporting using generative AI in at least one business function.

Despite their impressive performance, foundational models face critical limitations in domains where factually grounded, up-to-date information generation is essential. A phenomenon known as "hallucination"[44], in which models generate plausible but incorrect or fabricated outputs raises concerns about their reliability. These models are trained on static datasets bounded in time, and infer results with a limited context window. As a result, they often underperform on domain-specific tasks, especially when reasoning over complex information[63].

These shortcomings served as the primary incentive for the development of retrieval-augmented methods designed to ground model outputs in external, verifiable knowledge. Tasks automating decision-making in fields such as healthcare, law, and finance require not only linguistic fluency but also factual accuracy and traceability, which standalone generative models struggle to provide.

## 1.2 The Rise of Retrieval-Augmented Generation (RAG)

To mitigate the limitations of standalone generative models, the Retrieval-Augmented Generation (RAG) framework was introduced by Facebook AI Research (FAIR) team in 2020.[40] RAG fundamentally alters the generative process by integrating an external knowledge source commonly known as a "corpus", directly into the pipeline. The core methodology, though implemented with variations, remains consistent across a growing ecosystem of RAG systems. It consists of a three-stage process: retrieve, augment, and generate. The figure 1.1 shows an overview of the approach of the FAIR team. First, a retriever module encodes a user's query into a vector representation and uses it to search a pre-indexed corpus for the most relevant information chunks, which can be a dynamic top-k amount. The most relevant chunks are identified via a similarity score established between the query and chunks' vector representations. Second, these retrieved chunks are concatenated with the original query to form an augmented prompt. Finally, a generator model produces a response conditioned on this augmented prompt, which is now rich with relevant, external context.



**Figure 1.1:** Overview of the FAIR RAG pipeline introduced in their paper[40]. They combine a pre-trained retriever and a pre-trained generator, fine-tuning them end-to-end on a downstream task.

This architecture offers several distinct advantages over non-augmented models. The FAIR team demonstrated it improves the factual accuracy and specificity of generated responses by grounding them in source material.[40] It also introduces traceability, as the system can cite the specific documents used to generate an answer, a critical feature for verification and trust.

The utility of RAG frameworks extends beyond reducing hallucinations and addressing limitations in generalisation. It also enables a foundational model to stay relevant over time without additional costly training[28], as up-to-date or new domain-specific information can be continuously integrated into the knowledge base. Recent developments in domain-specific RAG pipelines have sparked interest in the healthcare industry to assist medical professionals, although serious limitations are highlighted[64]. SentimentCareBot[49], a RAG Chatbot for mental health support may help address the growing demand for accessible specialist support, such as psychologists and counsellors.

By decoupling a model's innate reasoning capabilities from its stored knowledge, RAG

### 1.3. THE CHALLENGE OF HETEROGENEOUS DATA AND THE EMERGENCE OF EXPERT SYSTEMS

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represents a pivotal architectural shift to create systems that are more dynamic, reliable, and adaptable.

## 1.3 The Challenge of Heterogeneous Data and the Emergence of Expert Systems

While the initial RAG paradigm proved highly effective for text-based corpora, its monolithic approach revealed limitations when confronted with the complexity of real-world information environments. Corporate and academic knowledge bases are rarely composed of clean, unstructured text alone; they are heterogeneous collections of text, tables, charts, scanned forms, and diagrams with complex visual layouts.

As we further develop in Chapter 4, in our consulting use case, reports may include a mix of dense textual analysis, statistical tables, and infographics, such as the one shown in Figure 1.2 below.

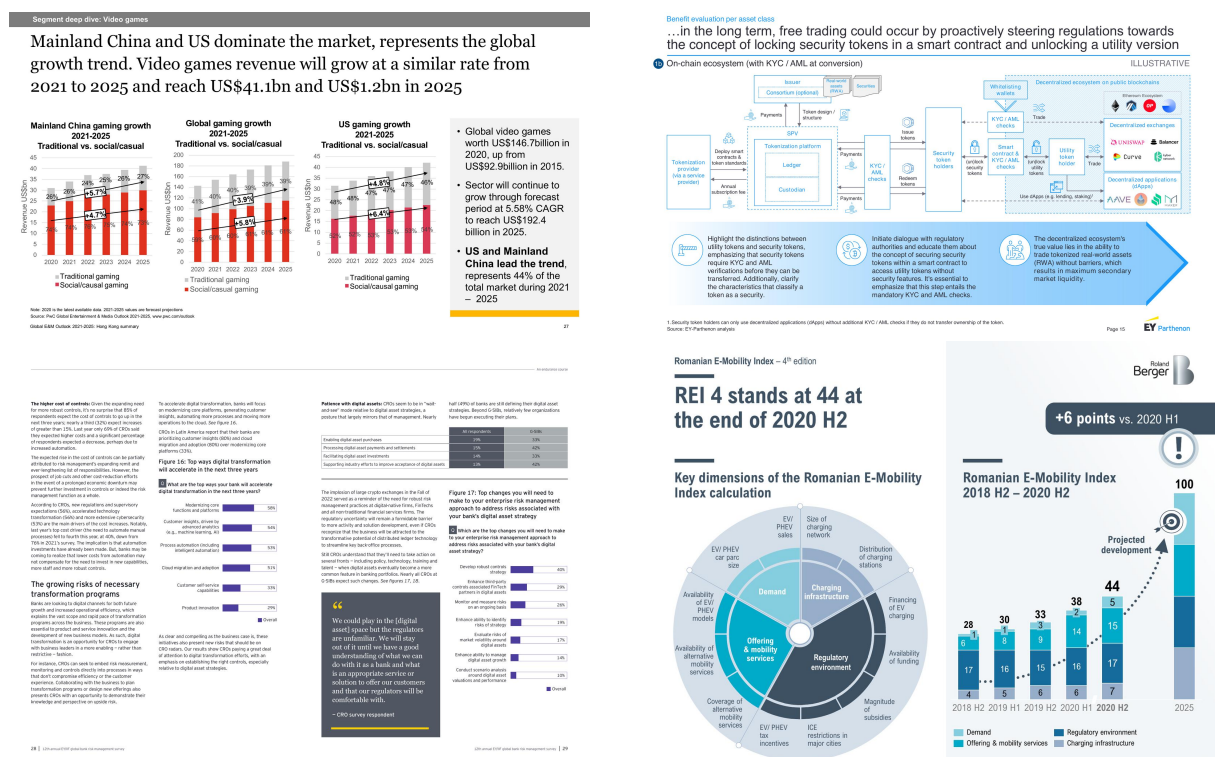


Figure 1.2: Example of a complex documents with mixed modalities.

A unimodal, text-centric RAG system struggles to interpret such content, as critical semantic information is often embedded in the visual structure rather than the raw character stream.

This challenge has driven the evolution of RAG away from a monolithic paradigm and toward a set of specialised, "expert" systems, each tailored to a specific data type or reasoning task. This divergence reflects a broader trend in artificial intelligence, moving

from singular, all-encompassing models toward more modular and composable architectures. For instance, multimodal RAG frameworks, such as ColPali, have emerged to handle visually-rich content directly, bypassing brittle text-extraction pipelines by generating embeddings from document images.[32] Concurrently, Graph RAG systems such as PathRAG[22] and LightRAG[36] were developed to address the need for multi-hop, relational reasoning by modelling explicit connections between entities in a knowledge graph. Another emerging less common modality is Audio RAG, where spoken interactions (e.g. call centre recordings or interviews) are indexed directly.[24] If naively transcribed, the information about the tone, emphasis, or hesitation that are critical for sentiment analysis and intent detection is lost.

The existence of these distinct, high-performing specialisations, some optimised for perception, others for reasoning, naturally poses the next critical research question: how can their complementary strengths be combined to create a system that is robust across the full spectrum of heterogeneous data?

## 1.4 Thesis Contributions and Structure

In a real-world setting, RAG systems must effectively integrate and retrieve information from diverse modalities, including text, images, sometimes even audio, or video content. This requires not only robust retrieval mechanisms but also sophisticated reasoning capabilities to synthesize information from multiple sources. In our case study, we take consulting as an example due to its high variety of data types, based on the company industry a consultant operates in, and the various tasks a consulting company achieves.

For instance, a consultant who may operate in a pharmaceutical company that requires extensive research and analysis of clinical trial data, might be focused on legal and safety reports, while when they operate in a food chain that needs insights into consumer behaviour and market trends, they might be looking at statistical and behavioural reports. Therefore the reports may include text-heavy reports, tables, charts, or images. The consultant must then synthesize this information to provide actionable recommendations to both clients.

This research project addresses the challenge of building a unified, high-performance RAG system for complex, real-world information environments. It makes two primary contributions to the field.

First, this work introduces MultiRAG, a novel and modular framework designed to aggregate and rerank candidate documents from multiple, independent "expert" retrieval systems for text and image modalities. By fusing the outputs of specialised textual and multimodal retrievers, MultiRAG leverages their complementary strengths to provide a more comprehensive and relevant context to the generator, improving overall system performance.

Second, this research proposes a new, LLM-driven methodology for generating domain-specific evaluation datasets. Recognizing the critical gap in existing benchmarks, which

often lack the complexity and heterogeneity of practical RAG deployments, this semi-automated process creates robust, multimodal evaluation corpora tailored to real-world use cases. This methodology facilitates more rigorous benchmarking and supports future research into advanced, retrieval-augmented systems.

The remainder of this thesis is structured as follows. Chapter 2 provides a comprehensive review of the state-of-the-art in RAG systems, from their foundational architectures to advanced specialisations. Chapter 3 details the technical architecture of the Multi-RAG framework and its hybrid fusion algorithms. Chapter 4 presents the novel dataset generation methodology. Chapter 5 describes the experimental setup and presents a thorough evaluation of the MultiRAG framework against strong baselines. Finally, Chapter 6 concludes with a summary of findings, a discussion of limitations, and directions for future work.

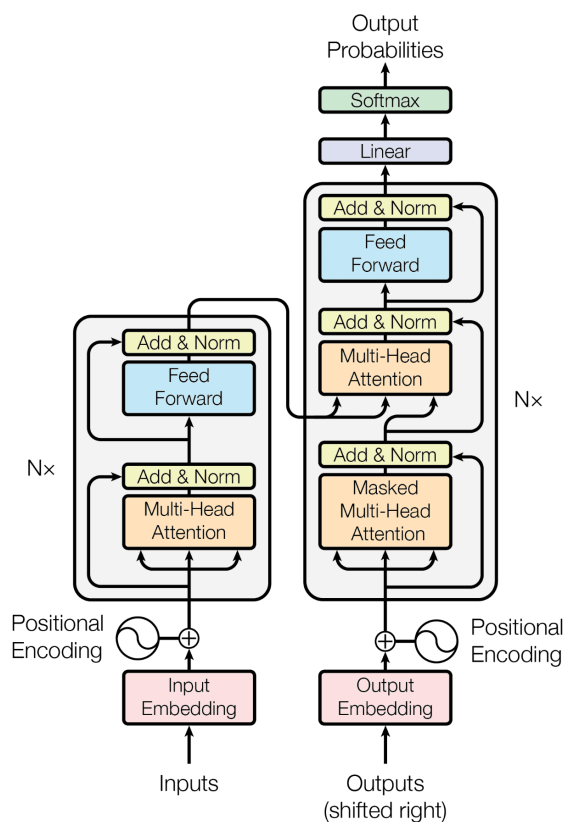
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# Chapter 2 – Literature review on State-of-the-Art in Retrieval-Augmented Systems

## 2.1 Foundational Architectures: The Transformer

The Transformer architecture is the technological bedrock upon which modern RAG systems are built.[60] Introduced in 2017, its design replaced the sequential processing of recurrent neural networks (RNNs)[55] with a parallelisable mechanism based on self-attention. This allows every token in an input sequence to contextualize itself relative to all other tokens simultaneously, enabling the efficient modeling of long-range dependencies.

At its core, the Transformer consists of an encoder and a decoder, each constructed from stacked layers of multi-head self-attention and positionwise feed-forward networks, as illustrated in Figure 2.1.



**Figure 2.1:** The Transformer architecture. On the left, the encoder processes the input sequence, while on the right, the decoder generates the output sequence.

**Self-Attention Mechanism:** The self-attention mechanism allows the model to weigh the relevance of different tokens when encoding a given token’s context. Each token is transformed into a query, key, and value vector. Attention scores are computed by taking the scaled dot-product between queries and keys, then using these scores to weight the corresponding values. This operation is performed across all positions in the sequence, allowing the model to encode dependencies regardless of their distance.

$$\text{Attention}(Q, K, V) = \text{softmax} \left( \frac{QK^T}{\sqrt{d_k}} \right) V$$

**Where:**

$$\begin{aligned} Q &\in \mathbb{R}^{n \times d_k} && : \text{matrix of queries} \\ K &\in \mathbb{R}^{m \times d_k} && : \text{matrix of keys} \\ V &\in \mathbb{R}^{m \times d_v} && : \text{matrix of values} \\ d_k &&& : \text{dimensionality of keys (used for scaling)} \\ n &&& : \text{number of query positions (tokens)} \\ m &&& : \text{number of key/value positions} \end{aligned}$$

**Multi-Head Attention:** Rather than applying a single attention function, Transformers perform multiple attention operations in parallel, each in a separate representation subspace. This is known as multi-head attention. It allows the model to jointly attend to information at different positions and representation types, capturing a richer set of dependencies across the input sequence.

**Positional Encoding:** Since Transformers do not have any inherent sense of sequence order (unlike RNNs), they incorporate positional encoding to inject information about the position of each token. These encodings, based on sinusoidal functions or learnable embeddings, are added to the input embeddings at the bottom of the encoder and decoder stacks.

**Feed-Forward Layers:** Each layer in the encoder and decoder also contains a positionwise feed-forward network, consisting of two linear transformations with a ReLU activation function in between. These layers operate independently on each position, helping the model capture nonlinear transformations of the input.

**Variants of Transformer Models:** Transformer-based architectures serve different purposes depending on how the encoder and decoder are used: Encoder-only models like BERT[29] excel at understanding tasks such as classification and sentence embeddings. Decoder-only models like GPT[21] are optimised for generative tasks, producing coherent language conditioned on prior context. Encoder-decoder models like T5[51] are designed for sequence-to-sequence tasks such as translation and summarization.

Extensions of the Transformer also include Vision Transformers (ViT)[30], which apply the same self-attention principles to image patches, and multimodal models like CLIP[50] and DALL·E[2], which align image and text representations through contrastive learning or generative objectives.

**Transformers in RAG Pipelines:** In Retrieval-Augmented Generation (RAG) frameworks, Transformer architectures serve a dual role. Encoders are used to embed queries and documents into a shared vector space for similarity-based retrieval. Meanwhile, generator models (typically decoder-only or encoder-decoder variants) synthesize final responses based on the query and the retrieved passages. Their capacity for integrating multiple contexts, learning from large-scale pretraining, and modeling long-range dependencies makes Transformers especially well-suited to retrieval-augmented reasoning.

## 2.2 Embeddings and Similarity Search

At the core of any retrieval system lies the similarity score, a function that quantifies the relevance of a document to a given query. The method of computing this score has evolved significantly, moving from straightforward vector comparisons to more nuanced, fine-grained interaction models.

- **Traditional Similarity Metrics for Single-Vector Representations:** In many dense retrieval models, both the query and the document are encoded into a single, high-dimensional vector. The similarity between them is then calculated using a simple and efficient metric.
  - **Dot Product:** This is one of the simplest and most efficient similarity measures. It is the sum of the element-wise products of the two vectors.

$$\text{similarity}(q, d) = q \cdot d = \sum_{i=1}^n q_i d_i$$

The dot product is sensitive to vector magnitude. This means that vectors with larger norms, which can result from longer documents or repeated term importance in some models, will tend to have higher similarity scores. This property can be desirable if vector magnitude is intended to encode a measure of importance.

- **Cosine Similarity:** To mitigate the influence of vector magnitude, cosine similarity measures the cosine of the angle between two vectors. It normalises the vectors, effectively judging them based purely on their orientation in the vector space.

$$\text{similarity}(q, d) = \cos(\theta) = \frac{q \cdot d}{\|q\| \|d\|} = \frac{\sum_{i=1}^n q_i d_i}{\sqrt{\sum_{i=1}^n q_i^2} \sqrt{\sum_{i=1}^n d_i^2}}$$

The score ranges from -1 (exactly opposite) to 1 (exactly the same), with 0 indicating orthogonality (no relation). It is particularly effective for textual data where document length can vary significantly without affecting the core topic. We decided to use cosine similarity as part of our MultiRAG framework for this reason.

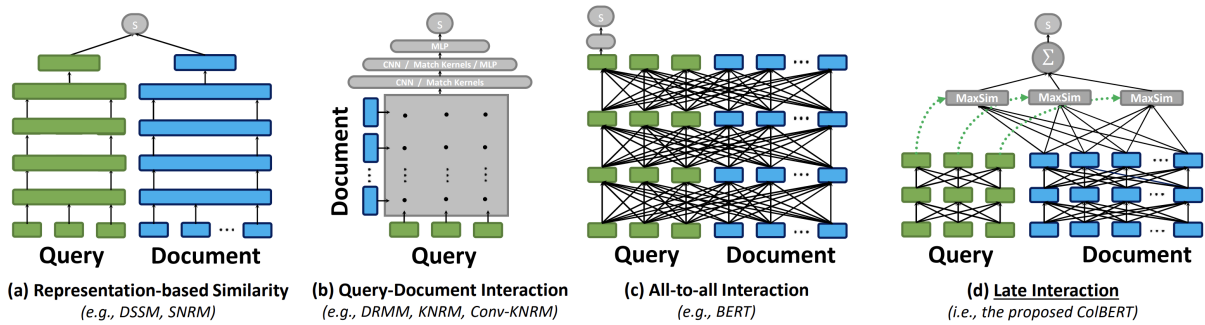
- **Euclidean Distance:** While the dot product and cosine similarity measure similarity, Euclidean distance measures *dissimilarity*. It is the straight-line distance between the two vector endpoints in the multidimensional space.

$$distance(q, d) = \|q - d\|_2 = \sqrt{\sum_{i=1}^n (q_i - d_i)^2}$$

A smaller distance implies greater similarity. To use it as a similarity score, it is often inverted or subtracted from a constant. For L2-normalised vectors, Euclidean distance and cosine similarity are monotonically related, but for unnormalised vectors, they capture different aspects of the vector relationship.

- **Late Interaction with Multi-Vector Representations:** While single-vector representations are efficient, they create an "informational bottleneck" by compressing the entire meaning of a document into one fixed-size vector. This can average out crucial details and fail to capture fine-grained term-level relevance.

Hybrid encoders now go beyond concatenating dense and sparse signals. Techniques inspired by ColBERT and FLMR architectures apply multi-vector representations where both documents and queries are decomposed into token-level embeddings, enabling **late interaction scoring**[39]. This approach, showcased in Figure 2.2, defers the similarity calculation, allowing for a more granular comparison between query and document terms at query time.



**Figure 2.2:** Late interaction scoring in ColBERT-style retrieval[39]. Contrasts existing approaches for query-document matching with the proposed late interaction paradigm.

The ColBERT (Contextualised Late Interaction over BERT) model exemplifies this paradigm:

- **Late Interaction Scoring (MaxSim):** The ColBERT scoring function computes document relevance via a late interaction mechanism. A document  $D$  is represented as a set of token embeddings,

$$D = \{d_1, d_2, \dots, d_n\},$$

and a query  $Q$  is similarly represented as

$$Q = \{q_1, q_2, \dots, q_m\}.$$

For each query token embedding  $q_i$ , the model finds its maximum similarity with any document token embedding  $d_j$  (typically using the dot product). These maximum similarity scores are then summed across all query tokens to produce the final relevance score:

$$Score(Q, D) = \sum_{i=1}^m \max_{1 \leq j \leq n} (q_i^T d_j)$$

This late interaction approach improves fine-grained relevance matching, particularly in retrieval-heavy tasks. It allows the model to identify multiple points of semantic overlap and rewards documents that match the specific contextual nuances of each query term, overcoming the limitations of a single, averaged vector representation. This method strikes a balance between the high effectiveness of full interaction models (like cross-encoders) and the efficiency required for large-scale retrieval.

## 2.3 Traditional RAG Pipelines

At the core of any text-centric RAG system lies a modular pipeline that orchestrates document preparation, retrieval, and generation. The process begins with document chunking, where raw texts (e.g. PDFs, HTML pages) are cleaned, normalised, and split into manageable passages.

Early approaches employed fixed-length character or token windows; however, modern implementations increasingly leverage **recursive splitting heuristics**[35][68] to avoid excessively small or large segments and **semantic chunking** via clustering embeddings to respect topical boundaries. Recursive splitters adjust boundaries based on punctuation or section markers, whereas **cluster-based semantic chunking** groups contiguous sentences by embedding similarity, yielding more coherent passages at the cost of increased computational overhead. Recent advancements also explore LLM-guided chunking, where language models are prompted to identify topic shifts or logical segment boundaries. The LumberChunker[31] paper, introduced at the ACL 2024, adaptively splits documents using a combination of language model predictions and perplexity scoring, significantly improving retrieval recall of specific information in a large corpus such as on the GutenQA benchmark. During this stage, metadata such as section headings and publication dates are also extracted to enhance downstream retrieval.

Next, each chunk is transformed into a vector representation in the **embedding generation** phase. Models such as Sentence-BERT [52], OpenAI’s text-embedding-ada [12], or domain-specific encoders encode semantic content into high-dimensional dense vectors. In parallel, **sparse representations** (e.g. BM25[53] term vectors) may be computed, and **hybrid embeddings** formed by concatenating dense and sparse signals, striking a balance between semantic understanding and keyword recall.

These embeddings are then indexed in a **vector store**, for example, FAISS [3], Weaviate [16], or Milvus [8], which exploits advanced data structures like inverted file systems

(IVF).4, product quantisation (PQ).5, and Hierarchical Navigable Small World graphs (HNSW).6 to support sub-linear nearest-neighbour search over millions of vectors. For simpler setups or experimental purposes, a local in-memory index may suffice, but these do not scale well.

During inference, the **retriever module** maps an input query into the same embedding space, retrieves the top- $k$  nearest chunks, and, when using hybrid methods, combines dense similarity scores with sparse term matches.

Finally, the **generator module**, typically a decoder-only model like GPT-4 or an encoder-decoder such as T5, conditions on the concatenation of the original query and retrieved passages to produce a fact-grounded response. Optionally, a **reranker** refines the retrieval results: cross-encoder rerankers rescore chunks by jointly processing query and passage, contrastive rerankers optimise embedding spaces for relevance discrimination, and in-context reranking leverages the LLM itself via few-shot prompts to prioritize the most pertinent evidence [26].

Retrieval optimisations include other simpler **query expansion**, automatically adding synonyms or contextually related terms via thesauri or small LLMs, and **dual-encoder architectures** that separately train query and document encoders to maximize retrieval fidelity. More recently, **contextual retrieval** techniques prefix or interleave document titles and pseudo-questions (generated by models such as HyDE), proposed at the NeurIPS 2022[34], which involves prompting an LLM to generate a synthetic "ideal answer", embedding it, and using it as a retrieval query. This closes the semantic gap between information need and raw input query, boosting needle recall and grounding quality.

Recent work such as the Hybrid Retrieval Benchmark (HyR[18]) presented at the ACL 2023, also introduces scoring functions that combine sparse-dense retrieval post hoc, through weighted fusion or learned reranking[43]. This hybridization remains effective across domains like legal, biomedical, and academic corpora.

Despite these advances, traditional RAG frameworks contend with several persistent challenges. Even when relevant passages are retrieved, LLMs may under-utilize them, reverting to memorized knowledge or hallucinating unsupported details, a phenomenon that undermines factuality. The quality of retrieval itself depends heavily on chunking strategy and embedding freshness; as corpora grow or drift over time, embeddings may become stale, necessitating frequent re-indexing. Scalability concerns loom large when indexing billions of tokens, as vector stores incur storage, latency, and cost overheads. Finally, most embedding models are pre-trained on general-purpose English text, resulting in suboptimal performance on specialised domains or non-English languages without additional fine-tuning or multilingual adaptation.

These limitations motivate exploration of complementary approaches, most notably graph-based and multimodal RAG, which aim to bolster multi-hop reasoning, cross-modal grounding, and dynamic knowledge integration.

## 2.4 Extending Perception with Multimodal RAG

Traditional RAG and Graph RAG architectures are predominantly unimodal, assuming that the input corpus is composed solely of text. However, many real-world documents, such as scanned PDFs, scientific papers, forms, and diagrams, contain complex visual layouts, embedded figures, or non-textual cues that textual embeddings alone cannot adequately capture. Multimodal RAG aims to extend the retrieval-augmented generation paradigm to such heterogeneous data, integrating visual and textual modalities to support retrieval, reasoning, and generation over richly formatted or image-based content.

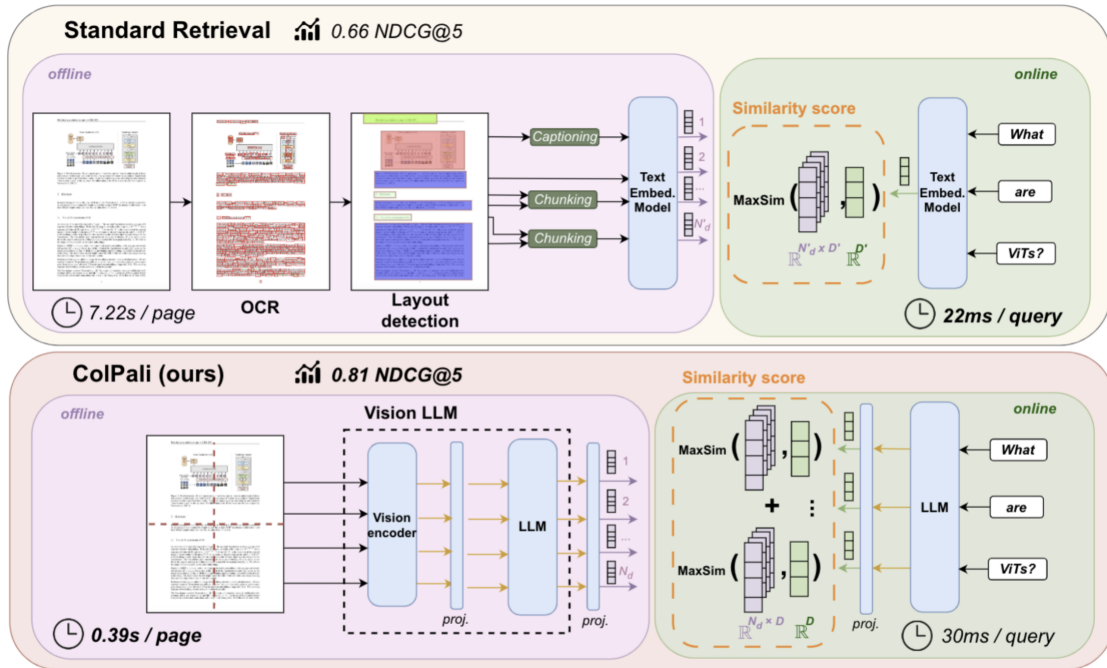
The earliest multimodal RAG systems rely on a pipeline that first applies optical character recognition (OCR) to extract text from images or scanned documents. This is typically followed by text chunking, layout parsing, and occasionally caption generation to describe figures or tables. While effective in some settings, this pipeline is slow, error-prone, and fragile when confronted with noisy inputs, such as poor-quality scans, handwritten text, or documents with intricate multi-column or nested layouts. Furthermore, semantic coherence can be lost during OCR, especially when visual cues like font size, colours, spatial alignment, or figure boundaries are discarded.

To address these limitations, newer systems incorporate vision-language models capable of learning joint representations over both images and text. PaliGemma [19], for instance, combines SIGLIP[69] (a vision transformer) with Pali-3[23] (a multilingual language model) to produce dense, cross-modal embeddings. This enables improved retrieval across image-text boundaries, allowing for more contextually aware and semantically grounded retrievals.

A more radical shift is embodied in ColPali[32], which bypasses OCR entirely. Instead of converting documents into text, ColPali treats each page as a raw image and encodes it directly using a multi-vector image embedding scheme with PaliGemma. These embeddings support late interaction retrieval, meaning that fine-grained relevance can be determined during inference rather than at index time. Late interaction retrieval means each query vector is compared to each image patch embedding and relevance is computed using a max-sum strategy, similar to ColBERT. This strategy enhances alignment between query semantics and document visuals, particularly for complex documents that defy easy text extraction.

Several emerging frameworks push the boundaries of multimodal retrieval-augmented generation. VDocRAG[58] supports multimodal reasoning by encoding rasterized documents into cross-modal vectors using contrastive pretraining. It also dynamically adjusts image resolution based on document complexity, allocating higher resolution to regions of interest, improving performance on visually dense layouts like tax forms and academic papers. It supports multi-hop reasoning across visual content.

ViDoRAG[61] takes a hybrid approach, combining textual and visual retrieval with a dynamic adjustment mechanism based on Gaussian Mixture Models (GMM), which learns weighted fusion across text and image modalities. It also introduces an agent-based generation framework, where specialised agents (e.g. seeker, inspector, answer agent)



**Figure 2.3:** ColPali differentiation[32]. It simplifies the existing document retrieval methods achieving better performances.

collaborate to process and synthesize multimodal evidence. This architecture, demonstrates strong performance in scenarios requiring cross-modal coordination and reasoning from disparate sources.

Similarly, VisRAG[66] adopts a query-centric retrieval strategy, retrieving multimodal evidence based on relevance to the query rather than document-centric similarity. Its modular design emphasises memory efficiency and component interchangeability, and it has shown competitive results on multiple visual question answering (VQA) tasks.

Finally, mmGraphRAG[17] introduces a dual-database architecture, combining a knowledge graph with a vector store, similarly to what has been done in HybridRAG[54]. It integrates object detection, spatial layout modeling, and image feature embeddings, creating a structured yet flexible framework for reasoning over visual documents. This design enables entity-centric, relational queries that span both textual and visual domains, unlocking richer and more compositional reasoning capabilities.

Despite recent advances, multimodal RAG systems face persistent challenges. OCR pipelines remain brittle and domain-specific, while vision-language models require extensive pretraining and careful calibration to ensure cross-modal alignment. The computational overhead associated with high-resolution image encoding, multi-vector matching, and agent-based generation can be substantial, limiting real-time applicability. Furthermore, the lack of standardised benchmarks and metrics for evaluating multimodal RAG performance across tasks, such as layout understanding, table reasoning, or figure caption grounding, complicates model development and comparison.

Promising directions for future research include the development of cross-modal knowl-

edge graphs, where visual objects, textual elements, and their relationships are explicitly represented and queried. Hybrid retrieval mechanisms that blend textual, visual, and structured representations may provide the flexibility needed for open-domain multimodal tasks. Another area of interest lies in streaming multimodal retrieval, where models adapt in real-time to evolving visual content (e.g. dynamic dashboards, live documents).

More broadly, multimodal RAG systems offer a path toward explainable AI[47], where retrieved evidence, whether textual, visual, or relational, can be explicitly surfaced and interpreted. This opens the door to interactive, agent-driven reasoning systems, capable not only of answering questions, but of demonstrating how and why a given response was generated. This is well demonstrated in ColPali with the similarity patches showing which patches of the images have high similarity with the query.

## 2.5 Unification over Mixture-of-Experts Retrieval

The prevailing paradigm in multimodal retrieval has largely favored unification: mapping all data modalities into a single, shared vector space to be queried by a single retrieval engine. This approach is underpinned by compelling arguments for its simplicity, and the promise of seamless cross-modal reasoning. The primary motivation is to avoid the significant architectural and operational burdens of a Mixture-of-Experts (MoE) system. Building and maintaining independent RAG pipelines for every modality such as text, images, and tables, and then attempting to merge their outputs post-retrieval may introduce high latency, complex orchestration, and the difficult problem of normalising inconsistent relevance scores across modalities.

A unified approach capitalizes on the synergy of shared or aligned embeddings. Foundational models like CLIP[50] and its successors have demonstrated the power of learning joint representations, enabling direct cross-modal retrieval. This allows a system to handle any-to-any queries (e.g. using a text query to find a relevant image, or an image to find related text passages) from a single index, simplifying the retrieval logic considerably. Proponents argue that a unified model can also better model cross-modal context, as its attention mechanisms can learn to focus on the most salient information regardless of its original modality. From a practical standpoint, it seems more efficient to align embeddings once and let a single, powerful engine handle retrieval across all data types.

However, this monolithic strategy is not without its trade-offs. While elegant, the unification approach forces a compromise. A single embedding model, even a powerful one, may not optimally represent the unique characteristics of every modality. This can lead to a versatile but not finely-tuned system, where the unified retriever performs adequately on average but is outperformed by specialised "expert" retrievers on tasks that depend heavily on a single modality. This performance gap highlights the central challenge addressed by our work: how to harness the strengths of specialised retrievers without succumbing to the complexity of ad-hoc fusion, thereby creating a system that is greater than the sum of its parts.

## 2.6 Needle-in-the-Haystack Problem

A major challenge for RAG is the Needle-in-the-Haystack scenario: retrieving a very small, critical piece of information called "needle" from a big, mostly irrelevant corpus, the "haystack". In this setting, even a strong average-recall system may fail entirely if the rare relevant chunk is not found. This problem critically affects grounding and faithfulness: if the needle isn't retrieved, the LLM may hallucinate or guess. Standard metrics (e.g. top- $k$  recall) often understate this risk, since missing one key fact can irreparably break an answer. One concrete illustration is the GutenQA benchmark introduced with LumberChunker[31] paper. It contains 3,000 QA pairs where each answer depends on a tiny fact in a long narrative. In such tests, specialised chunking and retrieval were needed to succeed. More broadly, several benchmarks emphasise this issue. For example, Sequential-NIAH[67] constructs contexts up to 128K tokens containing one or more "needles"; even the best models score only 63% accuracy extracting these items.

Similarly, the "Needle-in-the-Haystack" evaluation (e.g. Kamradt's LLMTest[9]) shows that as context size grows or multiple needles are added, retrieval performance degrades sharply.

Other open-domain QA tasks like HotpotQA[65] (distractor setting) and TriviaQA[38] also embody this: the answer may hinge on one sentence in a vast source. The implications for RAG are serious. Low recall on needles means RAG can produce confident but unsupported answers. It also complicates evaluation: a system might get an answer right by lucky inference, not retrieval. To mitigate this, recent research both creates new benchmarks and develops techniques to raise retrieval fidelity. For instance, methods like hypothesis-generation (HyDE) or chain-of-retrieval aim to boost the chance that needles are considered. Benchmark datasets like GutenQA and Sequential-NIAH explicitly test RAG systems on needle problems, highlighting how much room for improvement remains. Overall, the Needle-in-the-Haystack issue reminds us that RAG must not only retrieve relevant context on average, but must reliably capture the critical few facts; addressing this requires specialised indexing strategies, more exhaustive retrieval, and evaluation protocols that focus on rare-query sensitivity.

## 2.7 Evaluation Datasets and Benchmarks

The development of robust RAG systems is intrinsically linked to the quality and rigour of the benchmarks used for their evaluation. However, the current landscape of evaluation datasets reveals a significant gap, particularly for the complex, multimodal, and domain-specific scenarios that systems like MultiRAG are designed to address. Existing benchmarks, while valuable, often lack the specific challenges needed to test the limits of hybrid retrieval and fusion architectures.

Real-world datasets such as DocVQA[45] and its open-domain extension, OpenDocVQA[57], have been instrumental in advancing visual question answering over document images. Alongside these, synthetic datasets like ArxivQA[41], InfoVQA[46], TabfQuAD[32], or SlideVQA[56], used in benchmarks like ViDoRe[32] and ViDoSeek[62], have been cre-

ated to provide more controlled environments for evaluating multimodal information retrieval. These datasets have been incorporated into broader benchmarks, with ViDoRe now part of the Massive Text Embedding Benchmark (MTEB)[48], and many following the structure of the Benchmarking Information Retrieval (BEIR) framework[59] to standardise evaluation. This framework normalises datasets into three files: the "Corpus" lists the collection of documents to retrieve from, with unique ids. The "Queries" lists user-like query questions to be used, with unique ids. The "Qrels" standing for "Query Relevant Judgments" provides the ground truth for evaluation by mapping queries to relevant documents.

Although the standardization of practices has improved the consistency of evaluations, these resources still face several limitations. The generation of high-quality, visually-rich questions that require multi-hop reasoning or are context-independent remains a complex and costly endeavor.

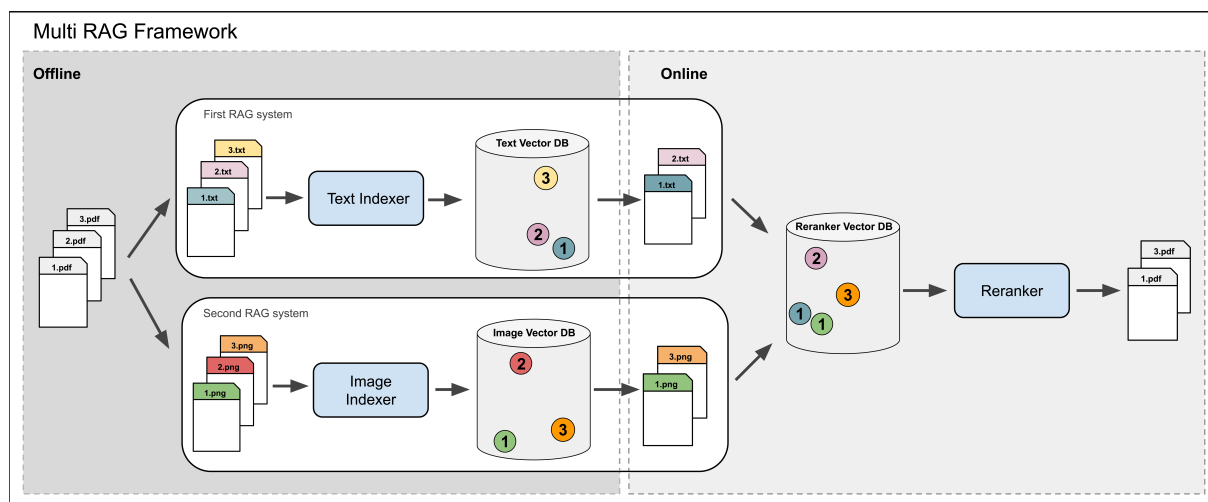
Furthermore, the models themselves can exhibit inherent biases towards certain modalities as a direct result of their training data. A model's architecture and fine-tuning regimen can optimise it for specific data types, which in turn can skew its performance when applied to tasks outside that domain as experimented by Cho et al.[25]. A pertinent example is the ColPali model, which serves as a baseline in this study. Its training was heavily centred on visually rich document benchmarks, including DocVQA, to enhance its performance on visual question-answering tasks.[32] This specialisation, while effective for its intended purpose, inherently biases the model towards excelling on documents with strong visual features over those that are purely text-heavy. Beyond such model-specific biases, annotation quality remains a significant concern, with factors like cost, inter-annotator consistency, and the required granularity of labels posing challenges. Coverage is also a critical issue, as most benchmarks are limited in domain specificity, language diversity, and the variety of modalities included.

In addition, these datasets are often cleaned of noise and are of a limited size, which prevents a realistic assessment of a system's performance at scale, especially concerning the Needle-in-the-Haystack problem where retrieving a single, vital piece of information from a vast corpus is challenging.

Looking forward, there is a clear and urgent need for a new generation of evaluation tools. This includes novel benchmarks explicitly designed to assess RAG workflows that ingest and synthesize information from multiple, heterogeneous modalities. To meet these needs, increased automation and sophisticated quality control mechanisms for dataset generation are essential to ensure diversity, precision, and annotation integrity at scale. It is this gap in the evaluation landscape that motivates the second major contribution of this thesis: a new, LLM-driven methodology for creating robust, domain-specific benchmarks tailored to the challenges of real-world RAG deployments.

# Chapter 3 – MultiRAG Framework and Hybrid Fusion Strategies

## 3.1 A Modular Architecture for Mixture-of-Experts Retrieval



**Figure 3.1:** High-level MultiRAG pipeline. The system consists of multiple independent retrieval pipelines, each optimised for a specific modality (e.g. text, images). The outputs from these "expert" retrievers are then aggregated and reranked in a final fusion step before being passed to the generator.

This work introduces MultiRAG, a modular framework designed to enhance retrieval performance over heterogeneous corpora by fusing the outputs of multiple, specialised retrieval systems. The high-level architecture, depicted in Figure 3.1, is predicated on a "Mixture-of-Experts" philosophy. Instead of attempting to build a single, monolithic index that handles all modalities equally well, MultiRAG employs independent, parallel retrieval pipelines, each optimised for a specific data type. The outputs from these "expert" retrievers, like a text-based system and a multimodal system showcased in this implementation, are then aggregated and intelligently reranked in a final fusion step.

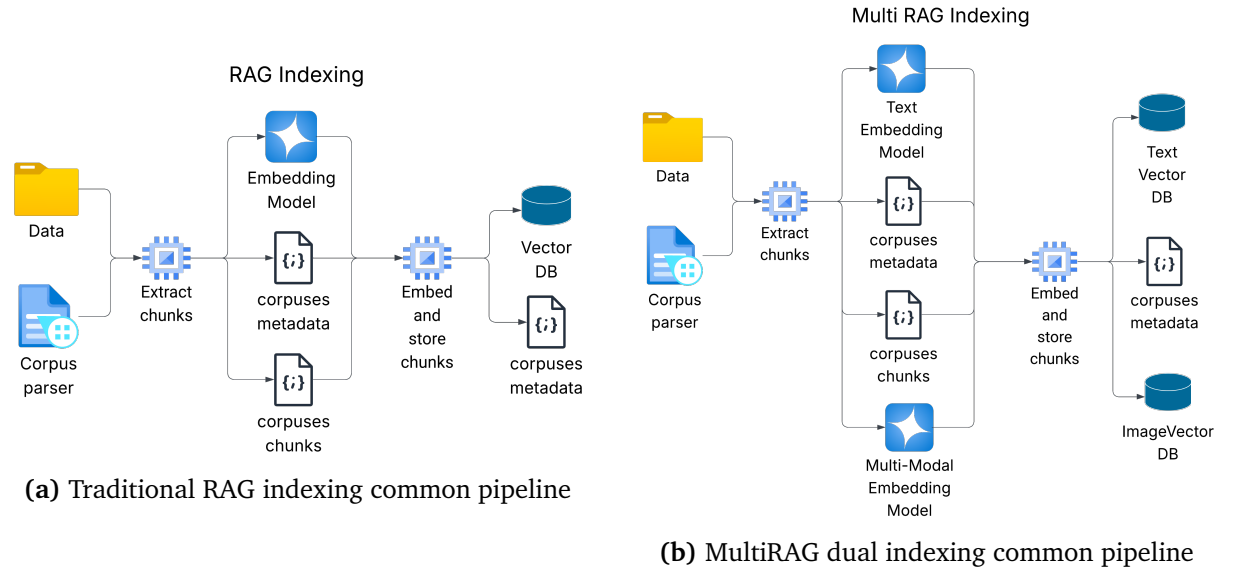
This design is motivated by the observation that no single retrieval method is universally optimal. Text-centric retrievers excel at keyword matching and semantic search over unstructured text, while vision-language models are superior at interpreting documents with complex layouts or embedded figures. By maintaining separate pipelines, each expert can be built using the best-in-class tools for its designated modality, avoiding the compromises inherent in a unified indexing approach. The core novelty of the framework lies in its final fusion stage, where the diverse signals from each expert are combined to produce a single, superior ranked list of candidate documents for the generator.

## 3.2 Implementation of Component Retrievers

To ground the abstract MultiRAG framework in a concrete implementation, this study constructs two expert retrieval pipelines that serve as the inputs to the fusion stage. These experts follow a common architecture consisting of document chunking, embedding generation, and vector indexing, but differ significantly in their treatment of input data and embedding strategies.

A generic overview of the indexing process for both original pipelines is shown in Figure 3.2a and the MultiRAG pipeline in Figure 3.2b.

Since the focus of this work is on the fusion strategies and their impact on retrieval performance, the generation phase is kept constant across all experiments, using the same LLM model for both traditional and multimodal RAG pipelines. Empirical qualitative results are compared in Chapter 5.



**Figure 3.2:** Indexing pipelines used in this work.

### 3.2.1 Traditional RAG Pipeline

The Traditional RAG Pipeline is optimised for text-centric documents. In this implementation, raw text is first extracted with commonly used libraries. For PDF documents, the PyMuPDF library is used to read the text content. For images, the PIL and Tesseract OCR engine are employed to convert the visual information into text. For raw txt or md files, no specific extraction is needed. The extracted text is then usually segmented into coherent passages using a semantic chunking strategy powered by a sentence-transformer model. However, for the purpose of this analysis, experimentation, benchmarking, and to maintain consistency across pipelines with our MultiRAG implementation, we decided to keep a full page as a chunk.

These chunks are then embedded using the nomic-ai/nomic-embed-text-v1 Hugging Face model[10], which is known for its strong performance, comparable to proprietary

models like OpenAI text-embedding-ada-002 and text-embedding-3-small, which remain industry-standard.

These dense vector representations are then indexed and stored in a PyTorch file, alongside their reference ids kept in a JSON Lines file. This local setup is chosen over a FAISS vector store for simplicity and ease of experimentation, given the relatively small size of the corpus used in this study.

During the retrieval phase, the input query is embedded using the same nomic-ai/nomic-embed-text-v1 model, and the top-k nearest chunks are retrieved based on cosine similarity. We chose  $k=100$  for computing all experiments, as it provides sufficient data points to perform an in-depth analysis of the retrieval performance across metrics. The metrics and results are discussed in Chapter 5.

Finally, for the generation phase, the top-k retrieved chunks are concatenated with the original query and passed to the qwen2.5vl model loaded through Ollama. This model is selected for its open-source availability, its ability to handle image inputs to fairly compare empirically its responses with multimodal RAG, and for its strong performance in generating coherent and contextually relevant responses.

### 3.2.2 Multimodal RAG Pipeline

The Multimodal RAG Pipeline is designed to handle visually rich documents and is based on the principles of the ColPali Framework.[32] Instead of text extraction, this pipeline treats each document page as a raw image. To adapt other file formats, PDFs are converted to images using the pdf2image library, while text files are rendered into images using the fpdf, PyMuPDF, and PIL libraries. The images generated from text files are cropped so that all text is visible with limited amount of white space from the original A4 format. Early on during this process, we experimented with rendering both full A4 pages versus cropped images, but found a negligible difference in score similarities between the two approaches, while the cropped images were slightly smaller in size therefore faster to process. For instance, for the MS-MARCO dataset, the size of a full A4 page is  $\sim 110\text{KB}$ , while the cropped version is  $\sim 95\text{KB}$ , and the mean score is 0.43 with a standard deviation of 0.08, while the cropped version is 0.44 with a standard deviation of 0.09. This was deemed acceptable for our purposes to ignore such differences. We therefore decided to use cropped images for all experiments. For original PNG or JPEG files, no conversion was needed, however, the images were resized to a maximum width of 1200 pixels to optimise RAM usage.

These images are directly encoded using the vidore/colpali-v1.3 Hugging Face model to generate multi-vector embeddings.[15] This approach captures both the textual content and the visual layout information of the page. Similarly to the traditional RAG, the resulting set of vectors for each page constituting a chunk is then stored locally for the experiments.

During the retrieval phase, the input query is embedded using the same vidore/colpali-v1.3 model, and the top-k nearest chunks are retrieved based on chamfer similarity score as in ColBERT mentioned in 2.3. Again,  $k=100$  is used for consistency across

experiments.

For the generation phase, the top-k retrieved chunks are concatenated with the original query and passed to the same qwen2.5vl model loaded through Ollama.

### 3.2.3 Multi RAG Pipeline

The MultiRAG pipeline builds upon the two expert retrievers described above. Therefore, the extraction, chunking, embedding, and indexing steps are identical to those in the Traditional RAG and Multimodal RAG pipelines. The key difference lies in the retrieval and fusion stages. It is therefore important to note that the memory and storage requirements for the MultiRAG pipeline are approximately double that of either individual pipeline, as two separate indexes must be maintained. However, this trade-off is justified by the potential for improved retrieval performance through the combination of complementary modalities, especially since storage is relatively inexpensive nowadays. Also the time taken to index the documents is roughly double, however, this is a one-time cost that can be amortised over many queries. With optimisations such as parallel processing and batching, the indexing time can be reduced significantly down to only slightly more than that of a single pipeline. For the purposes of this study, we did not implement such optimisations, as the indexing time was not a bottleneck.

MultiRAG combines the outputs of the Traditional RAG and Multimodal RAG pipelines. During the retrieval phase, both expert retrievers independently process the input query, each returning their top-k ( $k=100$ ) chunks based on their respective similarity metrics. These two sets of retrieved chunks are then merged via a reranker which plays a crucial role in sorting the context from both modalities into a single, coherent ranked list. The reranking algorithms explored in this study are detailed in the next section. We chose  $k=100$  for both pipelines to ensure a sufficiently large pool of candidates for the reranker to work with, while keeping the computational cost manageable. This high value of  $k$  allows to visualise the different scores across a large range of top-k chunks, which is important for our analysis.

The reranker outputs a final list of the top-k chunks, which are then concatenated with the original query and passed to the same qwen2.5vl model for generation. This ensures that the generation phase remains consistent across all three pipelines.

## 3.3 Multi-Retrieval Reranking

The central innovation of the MultiRAG framework is its fusion strategy, which combines the candidate document lists from each expert retriever into a single, optimised list. This study investigates several algorithms for this multi-retrieval reranking task, each embodying a different approach to aggregating evidence from heterogeneous sources. During inference, the top-k (e.g.  $k=100$ ) chunks from each pipeline are retrieved, duplicates are handled, and the combined set is then passed to one of the following fusion algorithms.

The reason the original scores from each retriever are not used directly is that they are often not comparable. Each expert retriever may use different scoring functions, resulting in scores that are on different scales or distributions. For example, a cosine similarity score from a text-based retriever may range from -1 to 1, while a BM25 score from a traditional information retrieval system can be unbounded and typically positive. A multimodal retriever may produce yet another set of scores with its own scale with the chamfer score. Directly merging these scores without normalisation can lead to biased results, where one retriever’s scores dominate the fusion process entirely.

We explore two options for normalising scores across retrievers: an embedding-based approach and a large language model (LLM)-based approach.

### 3.3.1 Reranking via Multimodal model embeddings

A multimodal embedding model such as Jina Embeddings V4[37] is used to embed both the query and each retrieved chunk from both pipelines. The cosine similarity between the query embedding and each chunk embedding is then computed, providing a unified score for all chunks regardless of their original modality. This approach leverages the model’s ability to understand both text and images, producing scores that are more directly comparable.

A challenge with this approach is the computational and storage cost, as it requires additional embedding computations for each chunk during inference, as well as increased memory usage to store the embeddings. However, this cost may be justified if it leads to improved retrieval performance. The system may be optimised to reduce inference bottleneck by precomputing embeddings for all chunks during indexing alongside other RAG pipelines used in MultiRAG. In addition, the choice of the multimodal embedding model is critical, as it must be capable of effectively capturing the semantics of both text and images to produce meaningful similarity scores, without biasing towards one modality. This can be difficult to achieve in practice, as many models are trained on datasets that are skewed towards one modality.

We specifically chose to use the jinaai/jina-embeddings-v4 Hugging Face model for this purpose[7], as it is designed to handle both text and images effectively.[37] The model architecture deliberately reduces the modality gap (the phenomenon where cross-modal pairs embed further apart than same-modal ones). Unlike CLIP-style dual-tower models, jina-embeddings-v4 uses a shared encoder which substantially improves alignment between text and image embeddings.

However, it is important to note that the effectiveness of this approach may vary depending on the specific characteristics of the datasets used, such as the balance between text-heavy and image-heavy documents, which is the reason for our in-depth analysis of reranking strategies.

### 3.3.2 Reranking via General Multimodal Model

General Multimodal Models (GMMs) such as Gemini[5] can be employed to re-evaluate the relevance of each chunk in the context of the query. This involves passing the query and each chunk through the GMM and using the model’s output as the final relevance score. This method can capture more nuanced relationships between the query and chunks, potentially improving retrieval performance.

The main drawback of this approach is the significant computational cost, as it requires multiple calls to the GMM during inference, which can be slow and expensive. The context size limit is also a major challenge, as the GMM may not be able to process all chunks if they are too large or if there are too many. This can be mitigated by limiting the number of chunks passed to the GMM, but this may also reduce the effectiveness of the reranking. In our analysis, we limited the number of chunks to 20 for this method to balance performance and cost.

Additionally, the performance of this method is highly dependent on the quality and capabilities of the GMM used. If the model is not well-suited to the specific retrieval task or domain, it may not provide meaningful relevance scores.

## 3.4 Fusion Algorithms

The choice of a fusion algorithm represents a critical trade-off between robustness and information fidelity. Since expert retrievers often produce scores on different, incomparable scales (e.g. cosine similarity versus BM25 scores), sophisticated methods attempt to mitigate this issue.

Score-based fusion methods retain the magnitude of relevance scores but require careful normalisation, which can be sensitive to outliers. In contrast, rank-based methods like Reciprocal Rank Fusion are highly robust to score distributions but discard potentially useful magnitude information. The following algorithms were implemented to empirically investigate this trade-off.

In our MultiRAG framework, we first attempt to normalise these heterogeneous scores to a common scale before applying any fusion method. Therefore, our experiments will showcase how well these normalisation strategies work in practice. If they are effective, we would expect score-based fusion methods to outperform rank-based methods. Conversely, if normalisation is poor, rank-based methods may prove more robust.

We chose Reciprocal Rank Fusion (RRF) as a baseline rank-based method due to proven robustness in reranking hybrid retrieval systems such as combining BM25 scores and cosine similarity scores. However, RRF has not been extensively tested in a multimodal retrieval setting, which is one of the contributions of this work. Additionally, we implemented score-based fusion methods that may optionally normalise scores from each retriever before fusing them.

### 3.4.1 Reciprocal Rank Fusion (RRF)

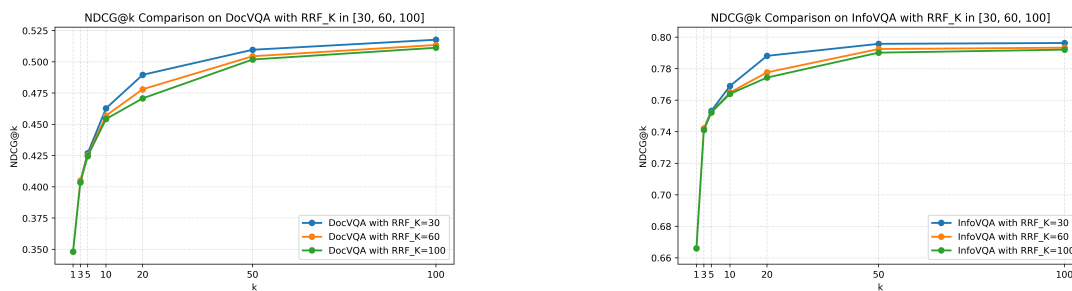
Reciprocal Rank Fusion (RRF)[27] is a rank-based method for combining the outputs of multiple retrieval systems. Instead of relying on raw retrieval scores, which may not be directly comparable across systems, RRF aggregates evidence using document ranks. Its reliance on ranks rather than raw scores makes it less sensitive to the specific scoring functions used by each retriever.

Formally, let  $\mathcal{R} = \{R_1, R_2, \dots, R_m\}$  be the set of  $m$  ranked lists returned by different retrieval modalities or systems. For a given document  $d$ , its rank in system  $R_j$  is denoted as  $\text{rank}_{R_j}(d)$ , where smaller values correspond to higher ranks. If  $d$  does not appear in  $R_j$ , its rank is set to  $+\infty$ .

The fused score of document  $d$  under RRF is defined as:

$$\text{RRF}(d) = \sum_{j=1}^m \frac{1}{k + \text{rank}_{R_j}(d)} \quad (3.1)$$

where  $k$  is a positive constant (commonly set between 50 and 100) that controls the contribution of lower-ranked documents. A higher  $k$  reduces the relative weight of high ranks, making the fusion less sensitive to small rank differences. In our experiments, we tested for  $k = 30$ ,  $k = 60$ , and  $k = 100$  across two datasets. The results in Figure 3.3 showcased  $k = 30$  to be the best parameter choice to balance the influence of top-ranked documents while still allowing lower-ranked ones to contribute, although the differences were marginal. In the final results presented in Chapter 5, we used  $k = 30$  and  $k = 100$  for all RRF experiments to demonstrate the importance of parameter selection.



**Figure 3.3:** Reciprocal Rank Fusion (RRF) for  $k$  in 30, 60, 100. The results show that  $k=30$  provides the best balance.

Finally, documents are sorted in descending order of  $\text{RRF}(d)$  to produce the fused ranked list. This method is robust across heterogeneous retrieval systems because it does not require score normalisation and favours documents that are consistently ranked highly across modalities. The algorithm is summarised in Algorithm 1.

---

**Algorithm 1:** Reciprocal Rank Fusion (RRF) Across Modalities

---

**Input:**  $T$ : list of (metadata, rank) pairs from traditional retrieval; $M$ : list of (metadata, rank) pairs from multimodal retrieval; $k$ : fusion constant (e.g. 100)**Output:**  $F$ : fused list of (metadata, RRF score) pairs**Step 1: Build ID  $\rightarrow$  rank mappings****foreach** list  $L \in \{T, M\}$  **do**    **foreach** ( $md, rank$ ) in  $L$  **do**         $cid \leftarrow$  extract identifier from  $md$         store  $rank_L(cid) \leftarrow rank$ **Step 2: Compute RRF scores** $AllIDs \leftarrow$  union of identifiers from  $T$  and  $M$ **foreach**  $id \in AllIDs$  **do**     $score \leftarrow 0$     **foreach** retriever  $R \in \{T, M\}$  **do**        **if**  $id \in R$  **then**             $score \leftarrow score + \frac{1}{k + rank_R(id)}$     Add ( $md, score$ ) to  $F$  with  $md$  metadata**Step 3: Sorting**Sort  $F$  in descending order of  $score$ **return**  $F$ 

---

### 3.4.2 normalise, Average and Fuse

In contrast to RRF, which is rank-based, this method is score-based and aims to leverage the actual relevance scores produced by each retriever.

The intuition behind this approach is that each retriever provides a different perspective on relevance, and by normalising and averaging their scores, we can create a more balanced and comprehensive assessment of document relevance.

The scores from each retrieval list are independently normalised using either L1 or L2 normalisation. After normalisation, the scores for each unique document are averaged across the lists in which it appears. The final list is then sorted by this fused average score.

The assumption is that the information contained across modalities is complementary, and averaging helps to mitigate the biases or weaknesses of any single modality. For instance, a document that scores moderately well across both text and image modalities may be more relevant than one that scores very highly in one modality but poorly in another.

This assumption might lead to poor performance if one modality is significantly more important than the other for a given query. However, in many practical scenarios, especially with heterogeneous data, this averaging approach can yield robust results. The

algorithm is summarised in the following Algorithm 2.

---

**Algorithm 2:** normalise and Average Scores Across Modalities
 

---

**Input:**  $T$ : list of (metadata, score) pairs from traditional retrieval;

$M$ : list of (metadata, score) pairs from multimodal retrieval;

norm: normalisation method  $\in \{L1, L2\}$

**Output:**  $F$ : list of (metadata, fused score) pairs sorted by score

**Step 1: Build ID  $\rightarrow$  score mappings**

$Acc \leftarrow$  an empty map from identifier to a list of (metadata, score) pairs

**foreach**  $(md, sc)$  in  $concatenate(T, M)$  **do**

$cid \leftarrow$  Construct unique identifier from  $(corpus\_id, doc\_id)$

    Append  $(md, sc)$  to  $Acc[cid]$

**Step 2: normalise scores per modality**

**foreach** score vector  $S \in \{T, M\}$  **do**

**if** norm = L1 **then**

$S \leftarrow S / (\sum S + \varepsilon)$

**else**

$S \leftarrow S / (\|S\|_2 + \varepsilon)$

    Replace original scores with normalised values

**Step 3: Compute fused scores**

$All\_IDs \leftarrow$  union of identifiers from  $T$  and  $M$

**foreach**  $id \in All\_IDs$  **do**

$vals \leftarrow$  normalised scores from available modalities

$md \leftarrow$  metadata from  $T$  if available, else from  $M$

**if**  $vals \neq \emptyset$  **then**

$fused\_score \leftarrow mean(vals)$

        Annotate  $md.fusion\_type \leftarrow$  **if**  $id \in T$  and  $id \in M$  **then**

            “both”

**else**

**if**  $id \in T$  **then**

                “traditional”

**else**

                “multimodal”

        Annotate  $md.fusion\_method \leftarrow$  “normalise\_average”

        Add  $(md, fused\_score)$  to  $F$

**Step 4: Sorting**

Sort  $F$  in descending order by fused score

**return**  $F$

---

### 3.4.3 normalise, Rerank and Fuse

Another score-based method combines the strengths of score normalisation with a rank-based fusion strategy.

First, as in the previous method, the scores from each retrieval list are normalised. The lists are then sorted based on these new normalised scores.

The final fused list is constructed using a "round-robin" approach, iterating through the ranks and adding the highest-scoring available document from either list at each step, while avoiding duplicates. This approach aims to balance the contributions from each expert retriever.

The intuition is that by normalising scores first, we ensure that each modality's relevance assessments are on a more comparable scale if the underlying distributions differ. The round-robin selection then allows us to interleave results from both modalities, ensuring that high-quality documents from either source are considered. The algorithm is summarised in Algorithm 3.

### 3.4.4 Simple Rerank and Fuse

This approach provides a simple method for merging results. It accumulates all retrieved documents from the traditional and multimodal pipelines into a single list. For documents retrieved by both systems, the one with the higher raw score is retained. The final list is then sorted in descending order by score. While straightforward, this method is susceptible to inconsistencies if the scoring scales of the underlying retrievers are not comparable.

Therefore, this is a good candidate to demonstrate the power of an effective multimodal reranker. If the embedding model used effectively embeds various modalities with limited bias, this strategy should outperform all others since it is always choosing the best chunks, no matter the modality, without smoothing different chunks retrieved with additional normalisation. The algorithm is detailed in Algorithm 4

The results for each algorithm described in this chapter are presented in the evaluation section 5.3.1, alongside an in-depth ablation study across all systems experimented.

---

**Algorithm 3:** normalise, Rerank and Fuse Across Modalities

---

**Input:**  $T$ : list of (metadata, score) pairs from traditional retrieval; $M$ : list of (metadata, score) pairs from multimodal retrieval;norm: normalisation method  $\in \{L1, L2\}$ **Output:**  $F$ : fused list of (metadata, normalised score) pairs**Step 1: Normalisation****foreach** modality list  $L \in \{T, M\}$  **do**    Extract score vector  $S$     **if** norm = L1 **then**         $S \leftarrow S / (\sum S + \varepsilon)$     **else**         $S \leftarrow S / (\|S\|_2 + \varepsilon)$     Replace scores in  $L$  with normalised  $S$ **Step 2: Sorting**Sort  $T$  and  $M$  in descending order by normalised score**Step 3: Fusion by Round-Robin Reranking**Initialize  $F \leftarrow \emptyset$ ,  $Seen \leftarrow \emptyset$ **for**  $i \leftarrow 0$  **to**  $\max(|T|, |M|) - 1$  **do**    Candidates  $\leftarrow \{T[i], M[i]\}$  if available

Sort Candidates by score descending

**foreach**  $(md, sc)$  in Candidates **do**        Construct  $cid$  from  $md$         **if**  $cid \notin Seen$  **then**            Add  $(md, sc)$  to  $F$             Mark  $cid$  as seen            Annotate  $md.fusion\_method \leftarrow \text{"rerank\_fuse"}$ **return**  $F$ 

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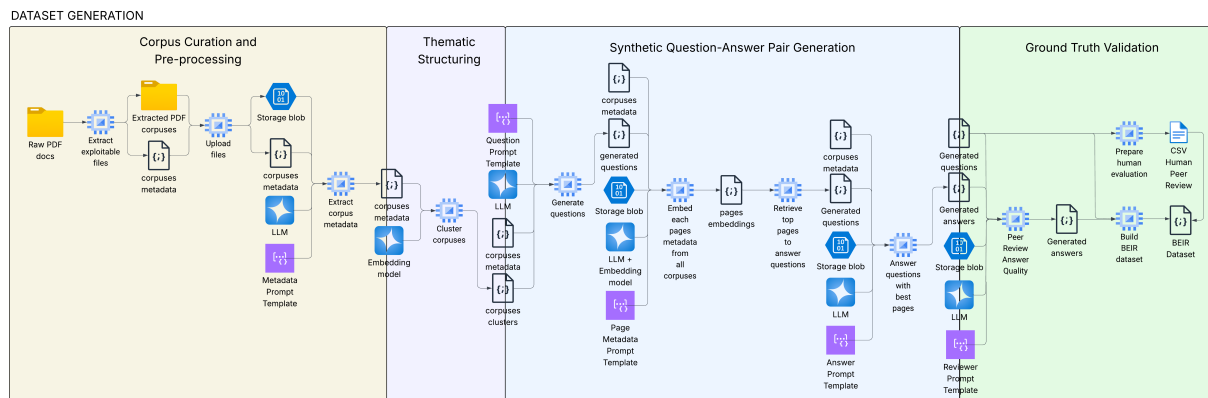
**Algorithm 4:** Merge by Maximum Score Across Modalities

---

**Input:**  $T$ : list of (metadata, score) pairs from traditional retrieval; $M$ : list of (metadata, score) pairs from multimodal retrieval**Output:**  $F$ : list of (metadata, score) pairs merged and sorted by score**Step 1: Initialize accumulator** $Acc \leftarrow \emptyset$  (map from identifier  $\rightarrow$  (metadata, score))**Step 2: Process each modality****foreach** ( $label, L$ ) in  $\{("traditional", T), ("multimodal", M)\}$  **do**    **foreach** ( $md, sc$ ) in  $L$  **do**         $cid \leftarrow$  extract identifier from  $md$         **if**  $cid$  is invalid **then**             $\hookrightarrow$  continue        Set  $md.source\_rag \leftarrow label$         **if**  $cid \in Acc$  **then**            **if**  $sc > Acc[cid].score$  **then**                 $\hookrightarrow Acc[cid] \leftarrow (md, sc)$         **else**             $\hookrightarrow Acc[cid] \leftarrow (md, sc)$ **Step 3: Sorting** $F \leftarrow$  values of  $Acc$  sorted in descending order by score**return**  $F$ 

---

# Chapter 4 – Methodology for Generating Domain-Specific Evaluation Datasets



**Figure 4.1:** Overview of the domain-specific dataset generation pipeline. The process is divided into four main stages. Each stage builds upon the outputs of the previous one, culminating in a high-quality, BEIR-formatted evaluation dataset.

## 4.1 The Imperative for Specialised Benchmarks

The robust evaluation of RAG systems is fundamentally constrained by the quality and relevance of the benchmarks used. While standardised evaluation frameworks such as BEIR and the MTEB have been instrumental in promoting reproducible research, they often fall short of capturing the nuances of specialised, real-world information environments, as discussed in Chapter 2. These general-purpose benchmarks frequently lack the specific mixture of heterogeneous modalities and complex, multi-step reasoning queries that characterise professional domains like consulting, finance, or law.

This gap is particularly evident in the context of multimodal retrieval. The Visual Document Retrieval (ViDoRe) benchmark, for instance, was introduced to evaluate systems on visually rich documents, moving beyond purely text-centric tasks. However, the rapid advancement of vision-language models has led to a saturation of its initial version. For instance, state-of-the-art models now exceed 90% nDCG@5 on the ViDoRe V1 benchmark, limiting its ability to meaningfully differentiate top-performing systems. This saturation indicates that the underlying task of single-page visual question answering is becoming a well-understood problem.

As the field is advancing towards the next frontier of challenges, ViDoRe V2 introduces more difficult scenarios like long-form, cross-document, and multilingual queries to better mirror real-world retrieval complexity. In addition to the ViDoRe V1 dataset used

to demonstrate comparable metrics to our baseline ColPali framework, we use ViDoRe V2 datasets such as ESG reports, Economic reports, and Biological reports in this work.

Many existing benchmarks are susceptible to the "Needle-in-the-Haystack" problem, where the task is to retrieve a single, precise fact from a vast and noisy corpus. This scenario is ubiquitous in corporate knowledge bases, yet it is often under-represented in cleaned, academic datasets. A system's true utility in a professional setting is determined not just by its average performance, but by its reliability in retrieving these critical, often rare, pieces of information.

The proposed pipeline is designed to generate queries that demand tailored retrieval and reasoning capabilities. This methodology is not merely a means to create another dataset, but a contribution of a scalable and reproducible recipe for building the next generation of benchmarks required to rigorously evaluate advanced RAG systems like MultiRAG.

## 4.2 Architectural Blueprint of the Generation Pipeline

The proposed methodology is structured as a four-stage, semi-automated pipeline designed to transform a corpus of raw, domain-specific documents into a high-quality, BEIR-formatted evaluation dataset.

- **Corpus Curation and Pre-processing:** Raw documents are ingested, and a small LLM is used to extract structured metadata at both the document and page level, providing a semantic foundation for subsequent steps.
- **Thematic Structuring and Content Analysis:** The extracted metadata is used to thematically cluster the documents, ensuring that generated questions are topically focused and that multi-document queries are drawn from a coherent set of related materials.
- **Synthetic Question-Answer Pair Generation:** A powerful generator LLM formulates a diverse range of questions based on the document metadata and then synthesises factually grounded answers by consulting the original source pages.
- **Ground Truth Validation via LLM Peer Review:** A separate LLM agent acts as a "peer reviewer" to validate the factuality, answerability, and relevance of each generated QA pair, filtering out low-quality or erroneous examples.

This modular architecture allows for a scalable and adaptable process. By leveraging LLMs for metadata extraction, question formulation, and quality control, the pipeline significantly reduces the manual effort typically associated with high-quality benchmark creation, while incorporating a level of semantic understanding that surpasses purely programmatic approaches. The following sections provide a detailed technical exposition of each stage.

An overview of the entire pipeline is illustrated in Figure 4.1.

## 4.3 Stage 1: Corpus Curation and Pre-processing

### 4.3.1 Source Selection and Ingestion

The initial and most critical step in creating a domain-specific benchmark is the selection of a representative corpus. For this project, a collection of 468 publicly available reports from leading global consulting firms (e.g. McKinsey & Company, Deloitte, KPMG) was curated. This choice was deliberate, as these documents illustrate the challenge of heterogeneous data that MultiRAG is designed to address. Consulting reports are inherently multimodal as showcased earlier in Figure 1.2, containing a rich and complex mixture of:

- Dense textual analysis: In-depth discussions of market trends, strategic frameworks, and case studies.
- Quantitative data tables: Financial statements, market sizing data, and performance metrics.
- Complex visualisations: Bar charts, line graphs, pie charts, and scatter plots illustrating key findings.
- Infographics and diagrams: Process flows, value chain maps, and conceptual frameworks.

This natural blend of modalities provides an ideal testbed for a hybrid retrieval system, as accurately answering a query often requires synthesising information from both textual and visual elements. The raw documents were in PDF format, and a total of 16,952 pages were ingested.

### 4.3.2 Automated Metadata Extraction

To enable targeted and semantically aware question generation, it is necessary to first create a structured understanding of the corpus content. gpt-4o-mini[11], a small and efficient language model, was prompted to act as an automated "librarian". The detailed prompts can be found in the Appendix .2. This model processed each document page-by-page to generate structured JSON metadata. This process involved two levels of abstraction:

- The document-level metadata: Resulted in a high-level summary for each report, with primary themes (e.g. "Market Analysis," "Technology Transformation," "Consumer Behaviour"), and extracted key entities such as industries, regions, and dates.
- The page-level metadata: It provided a description of the content of each individual page. This was particularly crucial for non-textual elements. For example, a page containing a bar chart would receive metadata such as: "description": "Bar chart showing video games revenue growth in Mainland China and the US from 2021 to 2025, with projected figures in US\$ billions".

This metadata provides the necessary foundation for the thematic clustering and question generation stages that follow. It allows the pipeline to reason about the content

of the documents at a high level without needing to process the full text of the entire corpus at every step.

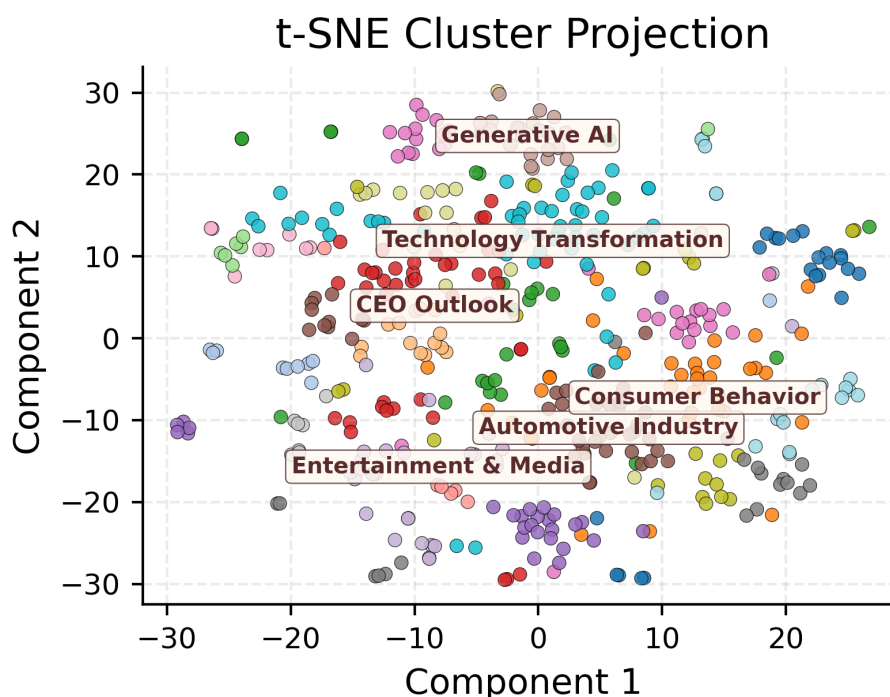
## 4.4 Stage 2: Thematic Structuring and Content Analysis

### 4.4.1 Topic Modelling via Semantic Clustering

With a rich metadata layer in place, the next stage organises the corpus into thematically coherent groups. This step is vital for ensuring that the generated questions are topically diverse and that multi-document questions draw from a logical and relevant set of source materials.

To partition the corpus into thematically coherent groups, each document's metadata was vectorised using the OpenAI text-embedding-3-small model. These embeddings were then grouped using a k-means clustering algorithm.

The crucial parameter, the number of clusters (k), was selected automatically. We computed the silhouette score for a range of k values and chose the one that yielded the maximum score, establishing an optimal k=31.



**Figure 4.2:** t-SNE visualisation of the thematic clusters derived from the document metadata. Each colour represents a distinct cluster, demonstrating clear thematic separation. Only a subset of clusters is annotated for clarity.

The resulting 31 clusters group documents by topic (e.g. "Generative AI", "Automotive Industry"), a structure leveraged in the next stage to generate contextually relevant

questions. As shown in Figure 4.2, a t-SNE visualisation confirms the distinctness of these thematic groupings.

## 4.5 Stage 3: Synthetic Question-Answer Pair Generation

This stage constitutes the core of the data generation process, where a powerful generative LLM (in this case, gpt-4) is employed to create high-quality, diverse question-answer pairs.

### 4.5.1 Taxonomy of Query Types for Consulting

To ensure the resulting benchmark robustly tests a range of reasoning capabilities, a taxonomy of six distinct question types was defined to reflect the analytical tasks common in a consulting context:

- **Recency-Bias:** Questions that test the model’s ability to identify time-sensitive relevant information when multiple sources conflict (e.g. “What recent shifts in strategic partnerships have been recommended for financial institutions to leverage blockchain opportunities?”).
- **Single-Fact:** Simple, direct questions requiring the extraction of a specific data point (e.g. “What is the expected growth rate of the warehouse automation market over the next five years?”). These test the system’s precision and its ability to handle the “Needle-in-the-Haystack” problem.
- **Single-Slide:** Questions that require understanding and synthesising the information presented on a single, often visually complex, page (e.g. “What are the projected top policy objectives to maintain US semiconductor design leadership?”). These test multimodal comprehension.
- **Multi-Slide:** Questions that necessitate connecting and comparing information across multiple pages or documents (e.g. “What strategic workforce planning and market adjustments are recommended for the automotive industry to mitigate employment impacts and supply chain risks from electrification and new tariff proposals?”). These test cross-document reasoning.
- **Aggregation:** Queries that require gathering multiple data points from across the corpus and performing an aggregation (e.g. “What are the common financial recovery indicators for hospitals across Southeast Asia and Japan following the COVID-19 pandemic?”). These test the system’s ability to synthesise information.
- **Top-Level Strategic:** High-level, open-ended questions that require inferring strategic insights from the provided materials, mimicking a consultant’s analytical task (e.g. “How should global luxury brands strategically align their digital marketing and promotional strategies to effectively engage young consumer segments in China’s evolving market?”).

## 4.5.2 LLM-Powered Question Formulation

For each thematic cluster, the gpt-4 model was prompted to generate questions corresponding to the taxonomy above. A key aspect of the prompting strategy is that the LLM was provided only with the metadata of one or more documents from a cluster, not their full content. To generate single document questions, the model received 1 document-level metadata entry, extracted randomly from the overall knowledge base preprocessed earlier on. For multi-document questions, the model was given the document-level metadata for 3 documents from the same cluster.

This approach forces the model to formulate questions based on the high-level concepts, structure, and described contents (e.g. charts, tables) of the documents, rather than simply extracting a sentence and rephrasing it as a question. This leads to more natural, abstractive, and less extractive queries which directly addresses one noted weakness of many existing benchmarks where queries are often just verbatim phrases from the source text.

The prompt also explicitly instructs the model to avoid overly generic or vague questions, and to ensure that each question is clearly answerable based on the described content. This helps maintain the quality and relevance of the generated queries.

For instance, the prompt for generating a recency-bias question might look like:

You are a domain expert tasked with designing a Recency Bias question based on the consulting document metadata provided. A Recency Bias question identifies insights that are temporally relevant, such as changes over time, trends, or recent recommendations.

Carefully read the document summary below and generate one insightful question that:

- References recent or time-specific insights (e.g. “in 2023”, “over the past 2 years”, “recently”).
- Highlights strategic shifts, trends, recommendations, or metrics that changed over time.
- Is clear, professional, and answerable using the content of the document.
- RETURN only the question as a sentence, nothing more.

Examples:

- ”What were the most recent strategic pivots recommended for mid-sized fintech firms facing regulatory changes?”

The document information:

topic: {topic}

```
summary: {summary}
```

The topic is a short phrase describing the main theme of the document while the summary is the detailed metadata description generated in Stage 1. The comprehensive list of prompts used for each question type is provided in the Appendix .2.

For our final Consulting dataset, a total of 300 questions were generated across the six defined types, 50 per type, ensuring a balanced representation of each category.

### 4.5.3 Context-Aware Answer Synthesis

Once a question was generated, a two-step process was initiated to produce a factually grounded, ground-truth answer:

- **Relevant Context Retrieval:** The generated question was embedded with the text-embedding-3-small model, and a vector similarity search was performed against the pre-computed embeddings of all page-level metadata descriptions in the corpus. The top-ranked pages were identified as the most likely sources needed to answer the question. For the single document questions, the top 3 most relevant pages from the same document were used to generate the question. For multi-document questions, the top 5 pages from the 3 documents were used to generate the question. This ensures that the answer is based on a focused and relevant subset of the documents, rather than the entire document or just their metadata.
- **Answer Generation and Citation:** The full content of these retrieved source pages, along with the original question, was then passed to the gpt-4 model. The model was instructed to synthesise a comprehensive and accurate answer based only on the provided context and to give a relevant and faithful response.

This process ensures that every generated answer is directly traceable to specific evidence within the source corpus, establishing a clear ground truth for evaluation. The tailored response to a limited amount of chunks made it easy to reproduce the BEIR format of queries, corpus, and qrels. Since each QA pair is linked to a unique corpus chunk in the commonly used format, question-answer pairs based on multiple documents were duplicated for each source chunk, with the same answer and question but different qrels pointing to the different chunks.

## 4.6 Stage 4: Ground Truth Validation via LLM Peer Review

In the literature, the final stage often involves human annotators to validate the quality of the generated QA pairs (e.g. DocVQA involved a pool of workers to annotate at different stages[45]). However, this approach is costly, time-consuming, and difficult to scale. Instead, we leverage the capabilities of a separate LLM instance to perform an automated peer review process.

### 4.6.1 The Reviewer Agent Protocol

To ensure the final dataset’s integrity and quality, a rigorous validation stage was implemented. A separate instance of gpt-4 was configured to act as an autonomous ”peer reviewer”. This reviewer agent was provided with the complete QA package: the generated question, the synthesised answer, and the full text of the cited source pages.

The agent was then tasked with performing a structured evaluation by answering a strict checklist of validation questions in a prompt (See Appendix .2), to augment commonly used generative metrics in a new final version of the answer:

- **Answerability:** Is the question fully and unambiguously answerable using only the information contained within the provided source pages?
- **Factuality and Faithfulness:** Is every claim or data point in the generated answer directly and verifiably supported by the provided context? Are there any hallucinations or unsupported inferences?
- **Conciseness and Relevance:** Is the answer directly responsive to the question without including extraneous or irrelevant information?

### 4.6.2 Quality Control and Dataset Finalisation

Although the reviewer agent is capable of nuanced judgement, a conservative approach was adopted to ensure high dataset quality. A subset of QA pairs was randomly sampled and manually inspected to verify the final QAs produced. We empirically judged and assessed the efficacy of this automated approach, which we further validate in the evaluation section amongst other baselines.

This four-stage process was used to generate the ”Consulting” dataset. The final dataset yielded 300 unique questions, which provisioned 1200 question-answer pairs pointing to a single document page. Since the original corpora contained 468 documents and 16,952 pages, this results in a sparsity of 0.07% in the BEIR format, highlighting the challenge of retrieving the correct page from a large and diverse corpus.

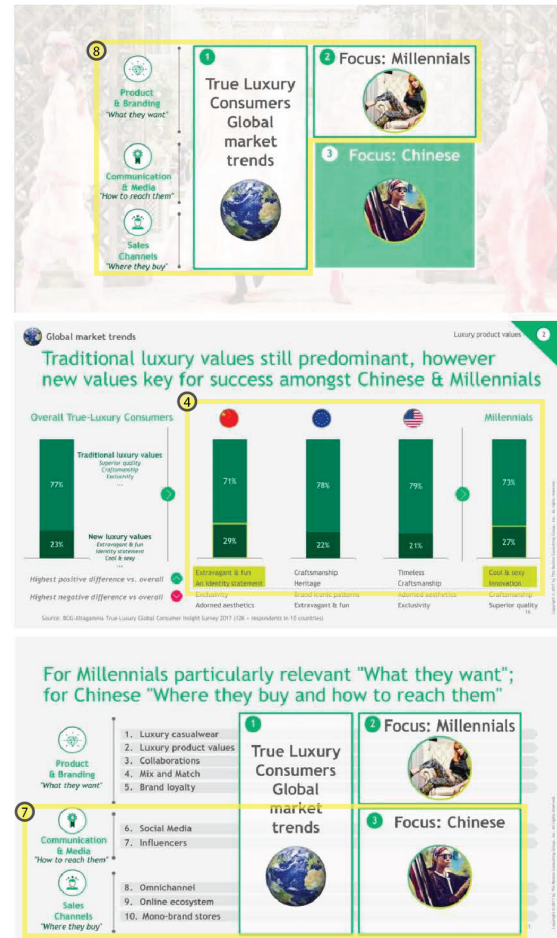
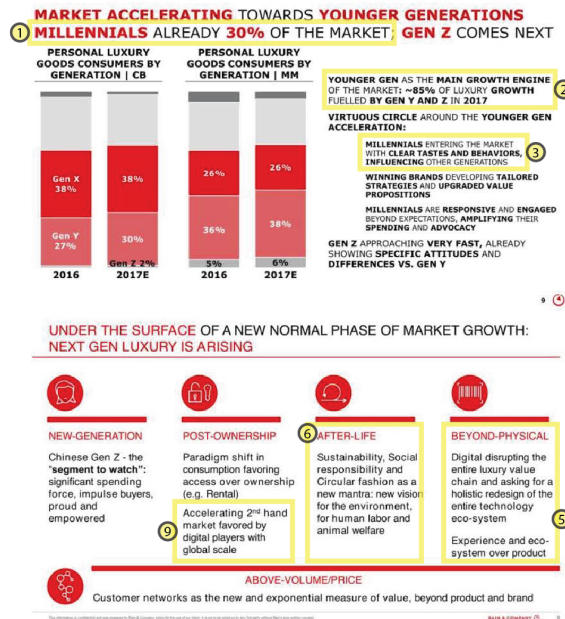
To fairly compare with existing benchmarks, a second version of the dataset was created by filtering out all document pages that were not cited in any QA pair, resulting in a smaller corpus of 1010 pages. This ”Consulting Light” dataset allows for a more direct comparison with benchmarks like ViDoRe V2, which also feature a limited number of documents, with each document serving at least one question.

An example of a generated question-answer pair from the final Consulting dataset is shown in Figure 4.3.

query-id: 162

query: How are luxury brand strategies evolving to cater to millennial and Gen Z consumers across different global markets while integrating technological and sustainability trends?

corpus:



answer:

#### Evolving Luxury Brand Strategies for Millennials and Gen Z

##### Overview and Market Trends

- Millennials represent 30% of the luxury market, with Gen Z emerging as influential consumers. Together, they drive approximately 85% of market growth, requiring brands to enhance value propositions.
- Gen Z consumers emphasize identity and experiential purchases, while millennials lead with clear preferences that influence broader market trends.

##### Strategies for Global Markets

- Personalization and Identity:** Brands focus on promoting products as identity statements, emphasizing "extravagant & fun" features along with traditional values like superior quality and craftsmanship.
- Technology Integration:** The digital ecosystem plays a critical role, supporting a holistic redesign across luxury value chains. This includes omnichannel strategies and influencer collaborations.
- Sustainability and Responsibility:** Key values include sustainability, social responsibility, and circular fashion, leading to increased preference for access over ownership and expanding secondhand markets.

##### Regional Adaptation

- Chinese Market Focus:** Strategies involve unique content on preferred sales channels, with emphasis on where and how to reach Chinese consumers.
- Western Markets:** Collaborations and mix-and-match approaches are pivotal, aiming to strengthen brand loyalty and extend reach through social media.

##### Implications and Next Steps

- Strategic Collaboration:** Brands should develop partnerships that elevate engagement with younger consumers through technology and personalization.
- Customer Experience:** Redefine customer journeys by focusing on ecosystems rather than isolated product offerings.
- Risks:** Balancing traditional and modern values might challenge brand identity coherence. Brands need strategic alignment with technological advancement and sustainability principles.

Figure 4.3: Example of a generated question-answer pair from the Consulting dataset. The question is designed to test multi-slide reasoning. We show the grounded answer with number pairs and highlighted boxes.

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# Chapter 5 – Evaluation and Results

## 5.1 Experimental Design and Objectives

This chapter presents a comprehensive empirical evaluation of the MultiRAG framework. The experimental design is structured to rigorously test the central hypotheses of this thesis and to quantify the performance of the proposed hybrid fusion strategies against strong, specialised baselines.

The evaluation is guided by three primary hypotheses:

H1: The MultiRAG framework, through its fusion of expert retrievers, will achieve at least as well as the best performing single expert retriever in retrieval performance. This hypothesis posits that by combining the complementary strengths of textual and multimodal retrievers, the fused system will be more robust and effective than either system operating in isolation. If a single expert is already optimal for a given query, meaning a single modality may be used to find the optimal documents, with the information not being captured by other expert systems, the fusion should not degrade performance and match the best single expert, thus ensuring that MultiRAG is a safe and beneficial enhancement. Since the datasets used in the evaluation contain a mix of document types, from text-heavy to visually rich, this hypothesis will be tested across a spectrum of scenarios to validate the generalisability of the MultiRAG approach, and its superiority in hybrid contexts.

H2: The effectiveness of different fusion algorithms will vary depending on the nature of the dataset, with rank-based methods like Reciprocal Rank Fusion (RRF) providing more robust performance in the face of non-normalised scores from heterogeneous experts. This hypothesis explores the nuances of the fusion stage, suggesting that methods which are less sensitive to the absolute score magnitudes of different retrieval systems will prove superior in a hybrid setting.

H3: The Consulting dataset generated through the proposed pipeline will present a challenging and realistic benchmark that effectively leverages multimodalities' strengths. Its large corpus size will showcase the limitations of academic datasets and highlight the practical advantages of the MultiRAG framework in real-world scenarios.

The following sections detail the evaluation protocol, including the datasets, metrics, and baselines used, and present a thorough analysis of the quantitative and qualitative results.

## 5.2 Evaluation Protocol

A standardised and rigorous evaluation protocol was established to ensure that the comparisons between systems are fair, reproducible, and meaningful.

### 5.2.1 Datasets and Benchmarks

To assess the generalisability and domain-specific performance of the MultiRAG framework, a diverse suite of 11 datasets was employed. This includes 10 established, public benchmarks and the novel, domain-specific "Consulting" dataset generated using the methodology described in Chapter 4. The datasets are grouped by their primary modality to structure the analysis:

- **Visually-Rich Benchmarks:** These datasets are directly drawn from the ViDoRe V1 collection[14]. They contain documents where visual layout, charts, and figures are critical for comprehension.
  - **ArxivQA:** Scientific question answering over ArXiv papers combining textual and visual (figure) content (500 corpuses, 500 queries, 500 QA pairs).
  - **DocVQA:** Visual document question answering over scanned and photographed documents (500 corpuses, 451 queries, 500 QA pairs).
  - **InfoVQA:** Question answering over infographics and charts requiring visual data interpretation (500 corpuses, 494 queries, 500 QA pairs).
  - **TabfQuAD:** Table-focused QA over financial and structured tabular data (70 corpuses, 280 queries, 280 QA pairs).
- **Text-Heavy Benchmarks:** These datasets are from the BEIR collection[1], they consist of dense, text-centric documents. Due to their large sizes, we decided to only use a subset of the datasets for efficiency when processing these with the Multimodal (image) retriever. We sampled a set of qrels for each dataset which was comparable to other datasets in size.
  - **MS-MARCO:** Large-scale real-world web search queries paired with relevant passages for open-domain QA and passage ranking (697 corpuses, 36 queries, 697 QA pairs).
  - **NFCorpus:** Consumer health question answering and retrieval over curated medical information documents (1000 corpuses, 48 queries, 1666 QA pairs).
  - **SciDocs:** Scientific document retrieval and citation-oriented recommendation over scholarly publications (1000 corpuses, 217 queries, 1069 QA pairs).
- **Other Domain-Specific Benchmarks:** Additional domain-specific datasets from the ViDoRe V2 collection[14] are more specialised in their content and structure.
  - **ESG Reports:** ESG-focused QA over sustainability and corporate responsibility reports with a food industry emphasis (1540 corpuses (from 30 unique documents), 228 queries, 888 QA pairs).
  - **Economics Reports:** Economic analysis and policy reports covering macro, trade and market themes (452 corpuses (from 5 unique documents), 232 queries, 3630 QA pairs).

- Biomedical Lectures: Biomedical lecture materials (slides + transcripts) for multi-modal educational QA (1020 corpuses (from 27 unique documents), 640 queries, 2060 QA pairs).
- Our Heterogeneous Benchmark:
  - Consulting Dataset: Public consulting industry reports from firm and consultant publications. This is the newly generated benchmark (Chapter 4), featuring a complex mix of text, tables, charts, and infographics, designed to be the primary testbed for real-world performance (16952 corpuses (from 468 unique documents), 300 queries, 1200 QA pairs).
  - Consulting Light Dataset: A filtered version of the Consulting dataset, retaining only documents that are cited in at least one QA pair, to allow for a more direct comparison with existing benchmarks (1010 corpuses (from 264 unique documents), 300 queries, 1200 QA pairs).

### 5.2.2 Performance Metrics

The evaluation focuses primarily on retrieval quality, as this is the core contribution of the MultiRAG framework. A robust evaluation of the retrieval phase of RAG systems requires targeted metrics, particularly in multimodal and reasoning-intensive settings. This study employs a concise yet comprehensive set of metrics to ensure meaningful assessment:

**normalised Discounted Cumulative Gain (nDCG)** evaluates the quality of ranked retrieval results by rewarding the correct placement of relevant documents near the top of the list. It accounts for the graded relevance of retrieved chunks and penalises lower-ranked relevant items, providing an effective measure of ranking quality in RAG pipelines.

$$DCG_p = \sum_{i=1}^p \frac{2^{rel_i} - 1}{\log_2(i + 1)}$$

$$IDCG_p = \sum_{i=1}^p \frac{2^{rel_i^{\text{ideal}}} - 1}{\log_2(i + 1)}$$

$$nDCG_p = \frac{DCG_p}{IDCG_p}$$

**Context Recall** measures the proportion of relevant evidence that was successfully retrieved. In high-stakes or imbalanced settings such as multimodal reasoning tasks, recall is critical, as it captures the model’s ability to detect all relevant content, minimising the risk of omitting essential information.

$$\text{Context Recall} = \frac{\text{Number of claims in reference supported by retrieved context}}{\text{Total number of claims in reference}}$$

**Context Precision** assesses the proportion of retrieved evidence that is truly relevant. It highlights how accurately the system avoids bringing in irrelevant or misleading documents. Precision is especially valuable in reducing noise in the retrieved context, which can negatively influence the generation phase.

$$\text{Precision@}k = \frac{|\{\text{retrieved docs}\} \cap \{\text{relevant docs}\}|}{k} = \frac{\text{True Positives@}k}{k}$$

**Mean Reciprocal Rank** is a metric that evaluates the effectiveness of a system in returning the first relevant item in a ranked list. For each query  $i$  in a set of queries  $Q$ , it calculates the reciprocal of the rank position (rank  $i$ ) of the first relevant item. The MRR is then the average of these reciprocal ranks across all queries. This metric is particularly useful in scenarios where the primary concern is retrieving at least one relevant item quickly, such as in question-answering systems or search engines.

$$\text{MRR} = \frac{1}{|Q|} \sum_{i=1}^{|Q|} \frac{1}{\text{rank}_i}$$

**Mean Average Precision** is a comprehensive metric that evaluates the precision of a retrieval system across multiple queries. For each query, it calculates the average precision (AP), which is the mean of the precision scores at each rank position where a relevant item is retrieved. The MAP is then the average of these AP scores across all queries. This metric is particularly useful in scenarios where it is important to consider the precision of all relevant items retrieved, not just the first one, making it suitable for evaluating systems that return ranked lists of documents or items.

$$\text{AP}_i = \frac{1}{m_i} \sum_{k=1}^n P_i(k) \cdot \text{rel}_i(k)$$

$$\text{MAP} = \frac{1}{|Q|} \sum_{i=1}^{|Q|} \text{AP}_i$$

Where  $m_i$  is the total number of relevant documents for query  $i$ ,  $n$  is the total number of retrieved documents,  $P_i(k)$  is the precision at rank  $k$ , and  $\text{rel}_i(k)$  is an indicator function that is 1 if the document at rank  $k$  is relevant and 0 otherwise.

Ultimately, no single metric can fully capture the nuanced performance of RAG systems, especially in multimodal and reasoning-intensive contexts. Therefore, the combination of these diverse metrics ensures a holistic view of system behaviour. While generation

quality is important, the generator model (qwen2.5vl) is held constant across all experiments. Therefore, generation metrics such as Faithfulness and Answer Relevance are reserved for a qualitative analysis which is out of scope for this work. Details can be found in the Appendix .3.

### 5.2.3 Baselines for Comparison

The MultiRAG framework is evaluated against its constituent expert components, creating a clear and direct comparison.

- **Baseline 1 (Text Expert):** The standalone Traditional RAG pipeline. This system uses text extraction followed by embedding with nomic-ai/nomic-embed-text-v1, a strong open-source text embedding model with a large context length. Retrieval is based on cosine similarity. We computed the scores of the system on a subset of datasets to establish which method performed best in between vector search only, BM25 only, and a hybrid of both. The hybrid method was found to be the best performing method on average across all datasets, and thus was used for the general evaluation across all datasets.
- **Baseline 2 (Visual Expert):** The standalone Multimodal RAG pipeline. This system is based on the ColPali framework and uses vidore/colpali-v1.3 to generate multi-vector embeddings directly from document page images. Retrieval is based on the late-interaction chamfer similarity score. Even though the model has received upgrades since v1.3, this version was used to ensure consistency with the original ColPali paper and to provide a strong, well-documented baseline.
- **Baseline 3 (Individual Experts with Reranking):** Each expert retriever is augmented with a reranking step. Two reranking strategies are employed separately for each expert:
  - Reranking with the jinaai/jina-embeddings-v4 model, a universal multimodal embedding model designed to reduce the modality gap between text and image representations.
  - Reranking with an industry-standard GMM model (Gemini 2.0 Flash).

This allows us to assess the impact of reranking on each individual expert system. In addition, the Jina model is also tested in ablation, directly embedding and retrieving the documents based on the queries, to see how it performs as a standalone retriever, both with textual and image inputs.

- **Proposed System (MultiRAG):** The full MultiRAG framework, which fuses the outputs of the Traditional and Multimodal pipelines. The reranking step is performed using Jina V4, as the ablation study shows its superiority to Gemini reranking. The framework is tested with each of the fusion algorithms detailed in Chapter 3.

The generative model for the generation phase is omitted during the evaluation since it isn't required to obtain the top k documents for each QA pairs of each dataset. This

extensive ablation is performed to isolate the impact of the retrieval phase, which is the core contribution of this thesis.

### 5.3 Quantitative Analysis of Retrieval Performance

This section presents the core empirical results of the evaluation, comparing the retrieval performance of the MultiRAG framework against the standalone expert baselines across the suite of benchmarks.

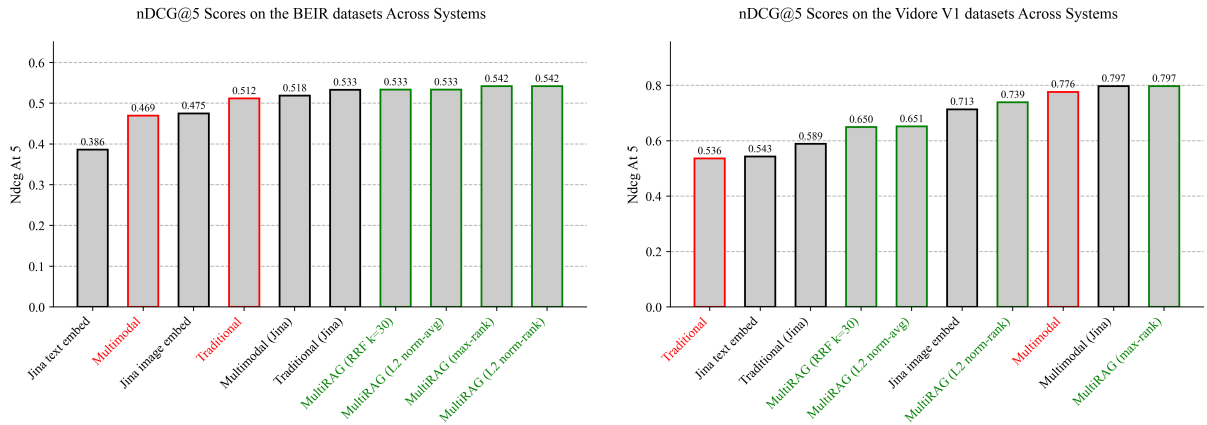
**Table 5.1:** Averaged retrieval results across existing benchmarks (ViDoRe V1, ViDoRe V2, and BEIR datasets) for each RAG system. The best scores for each metric are highlighted in bold.

| RAG system                    | nDCG@5       | Recall@5     | Precision@5  | MAP@5        | MRR@5        |
|-------------------------------|--------------|--------------|--------------|--------------|--------------|
| Traditional                   | 48.9%        | 42.4%        | 26.4%        | 35.9%        | 55.0%        |
| Multimodal                    | 66.2%        | 58.3%        | 32.3%        | 51.4%        | 73.8%        |
| Jina Text Standalone          | 46.9%        | 41.9%        | 24.6%        | 34.9%        | 52.0%        |
| Jina Image Standalone         | 57.5%        | 52.3%        | 28.1%        | 44.6%        | 65.9%        |
| Traditional (Jina Reranker)   | 58.7%        | 50.2%        | 31.4%        | 43.5%        | 63.3%        |
| Multimodal (Jina Reranker)    | 68.4%        | 60.1%        | 33.6%        | 53.0%        | <b>75.7%</b> |
| Traditional (Gemini Reranker) | 37.2%        | 34.9%        | 18.4%        | 27.3%        | 54.5%        |
| Multimodal (Gemini Reranker)  | 57.6%        | 53.0%        | 26.0%        | 44.9%        | 73.8%        |
| MultiRAG (rrf k=100)          | 61.6%        | 52.9%        | 32.2%        | 46.1%        | 68.7%        |
| MultiRAG (rrf k=30)           | 61.9%        | 53.4%        | 32.3%        | 46.4%        | 68.7%        |
| MultiRAG (L1 norm-avg)        | 61.9%        | 53.2%        | 32.2%        | 46.7%        | 69.0%        |
| MultiRAG (L2 norm-avg)        | 62.1%        | 53.2%        | 32.3%        | 46.8%        | 69.2%        |
| MultiRAG (L1 norm-rank)       | 65.2%        | 57.4%        | 33.0%        | 49.8%        | 72.3%        |
| MultiRAG (L2 norm-rank)       | 65.8%        | 57.9%        | 33.2%        | 50.3%        | 72.9%        |
| MultiRAG (max-rank)           | <b>68.6%</b> | <b>60.5%</b> | <b>34.0%</b> | <b>53.2%</b> | 75.5%        |

Table 5.1 summarises the primary results across the standard benchmarks, showing the scores for  $k$  retrieved documents set to 5 for comparable analysis with the academic literature such as ColPali. As expected, the expert retrievers Traditional and Multimodal RAG are consistently outperformed by our MultiRAG system.

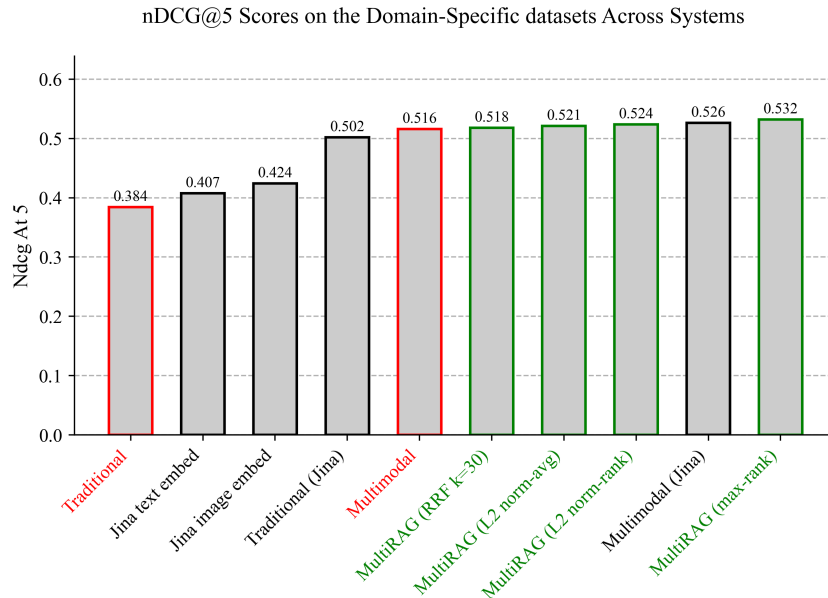
It is important to also demonstrate the performance of the systems on the individual dataset types, to show how they behave in different scenarios. In Figure 5.1, we show the results averaged for Text-Heavy Benchmarks (BEIR) separately from the Visually-rich Benchmarks (ViDoRe V1). These results demonstrate the performance of each system in different scenarios. Traditional RAG performs best on text-heavy datasets, while Multimodal RAG excels on visually-rich datasets. However, they are both outperformed by MultiRAG in all scenarios, which validates our first hypothesis (H1).

This indicates that the fusion of experts effectively leverages their complementary strengths, creating a system that is more robust and performs better on average across multiple



**Figure 5.1:** Retrieval performance (nDCG@5) on Text-Heavy BEIR Benchmarks (left) and Visually-Rich ViDoRe V1 Benchmarks (right) for each RAG system. MultiRAG (max-rank) consistently outperforms both baselines across all dataset categories.

data types. We confirm this further assessing the performance on the domain-specific datasets in Figure 5.2, including ViDoRe V2 and the "Consulting" dataset, which is discussed in more detail in section 5.3.2.



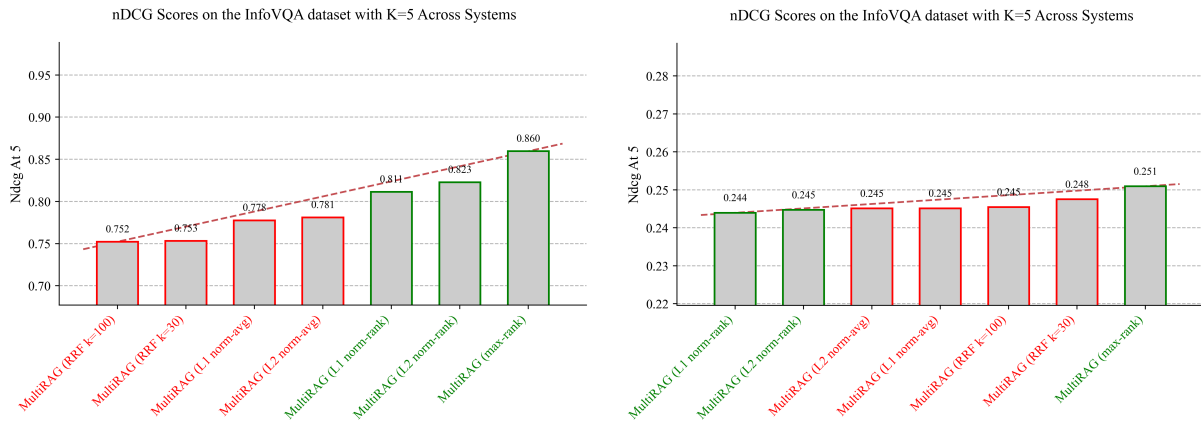
**Figure 5.2:** Retrieval performance (nDCG@5) on Domain-Specific Benchmarks (ViDoRe V2 and Consulting datasets) for each RAG system.

### 5.3.1 Ablation Study: The Impact of Fusion Algorithms

To test hypothesis H2 and understand the contribution of the fusion strategy, an ablation study was conducted on one heterogeneous and one homogeneous dataset in terms of modality leveraged. InfoVQA, a dataset considered highly heterogeneous in modality

since it only consists of infographics requiring visual interpretation. The Consulting dataset, on the other hand, is more homogeneous, consisting of text and infographics.

The two classes of reranking algorithm explored, besides their score-based or rank-based nature, are differentiated by their modality sensitivity. The commonly used Reciprocal Rerank Fusion (RRF) method described as Algorithm 1, as well as the normalised and average technique of Algorithm 2, are both favouring documents that are relevant across modalities. However, the other algorithms explored (normalisation and rerank (Algorithm 3), and maximum rerank 4) are favouring documents which are highly ranked by one expert, even if they are not ranked by the other expert.



**Figure 5.3:** Study of Fusion Strategies Strengths on the InfoVQA and the Consulting Dataset

The results of the study are summarised in Table 5.3. It is clear that the second class of methods outperform the first. It shows that in a heterogeneous setting, it is more important to capture the strengths of each expert, rather than trying to find documents that are relevant across modalities. This is likely due to the fact that in many cases, relevant documents may only be retrievable by one modality, and forcing agreement between experts may lead to missing these key documents. On a more balanced and homogeneous dataset such as Consulting, the difference between the methods is less pronounced, but the maximum rerank method still performs best. This validates our hypothesis H2, and shows that rank-based methods are more robust to the non-normalised scores from heterogeneous experts.

### 5.3.2 Performance on the Domain-Specific "Consulting" Dataset

The "Consulting" dataset serves as the ultimate testbed for the MultiRAG framework due to its complexity and the need for nuanced understanding across modalities.

To fairly assess the performance of this dataset against existing domain-specific datasets such as the ViDoRe V2 ones, a filtered version called "Consulting Light" was also evaluated, which includes only documents that are cited in at least one QA pair. This allows for a more direct comparison with existing benchmarks. However, this version comparable to academic dataset is lacking the scale of the full dataset, which is one of its key strengths. In industry settings, it is common to have large corpus of documents, many

of which may not be directly relevant to the queries at hand. The full "Consulting" dataset, with its 16952 chunks from 468 unique documents, attempts to close the gap with this reality and tests the scalability and robustness of the MultiRAG framework in a more practical scenario.

The results in Table 5.2 demonstrate both the complexity of the Consulting datasets and their homogeneous nature. It indicates a performance drop of  $\sim 10\%$  on for  $nDCG@5$  with a diminished standard deviation of 1%-2% between the traditional and multimodal baseline RAG systems, compared to the high average standard deviation of  $\sim 25\%$  observed for the ViDoRe V2 datasets, which supports our H3.

**Table 5.2:**  $nDCG@5$  retrieval results across domain-specific benchmarks (ViDoRe V2, and Consulting datasets) for each RAG system.

| RAG system                  | Esg Reports | Economics Reports | Biomedical Lectures | Consulting Light | Consulting |
|-----------------------------|-------------|-------------------|---------------------|------------------|------------|
| Traditional                 | 24.4%       | 25.2%             | 23.9%               | 37.4%            | 21.1%      |
| Multimodal                  | 53.7%       | 49%               | 56.2%               | 39.1%            | 22%        |
| Traditional (Jina Reranker) | 44.8%       | 50.2%             | 49.8%               | 39.3%            | 24.1%      |
| Multimodal (Jina Reranker)  | 46.8%       | 56.3%             | 58.7%               | 39.9%            | 23%        |
| MultiRAG (rrf $k=30$ )      | 46.3%       | 54%               | 52.2%               | 39.7%            | 24.8%      |
| MultiRAG (L2 norm-avg)      | 48.8%       | 52.6%             | 52%                 | 40.9%            | 24.5%      |
| MultiRAG (L2 norm-rank)     | 47.6%       | 51%               | 55.2%               | 41.3%            | 24.5%      |
| MultiRAG (L2 max-rank)      | 48.5%       | 51.8%             | 58.3%               | 41.4%            | 25.1%      |

In real-world applications, it might be sufficient to retrieve a whole PDF document, rather than a specific page. For instance, a library of thousands of books might be searched to find the most relevant book, and then the user can navigate within that book to find the specific information they need. This is a common use case in many industries, where documents are often long and complex, and users may not know exactly what they are looking for. This intuition led us to the creation of two scoring versions. The "v2" version is the commonly used page-level scoring (with a page being the equivalent of a chunk), where a retrieved page is considered relevant if it is matching the references of the QA pairs. The "v1" version we introduce, on the other hand, considers a retrieved page as relevant if it comes from a document that is cited in the reference answer.

This approach acknowledges that in many practical scenarios, retrieving the correct document is more important than pinpointing the exact page. We are interested in these two scoring versions to understand the performance of the system in its document comprehension capabilities compared to its difficulty in retrieving a needle in a haystack.

The scores in Table 5.3 demonstrate significant drops in performance for page-level retrieval:  $\sim 35\%$  for Consulting Light and  $\sim 40\%$  for Consulting. This supports the fact that the needle in the haystack problem remains a major challenge for advanced RAG systems have yet to overcome.

**Table 5.3:** nDCG@5 retrieval results across scoring versions of Consulting datasets for each RAG system.

| RAG system                  | Consulting Light V1 | Consulting Light V2 | Consulting V1 | Consulting V2 |
|-----------------------------|---------------------|---------------------|---------------|---------------|
| Traditional                 | 76.1%               | 41.9%               | 64.9%         | 24.5%         |
| Multimodal                  | 77.1%               | 43.2%               | 67.4%         | 25.5%         |
| Traditional (Jina Reranker) | 79.1%               | 44.0%               | 69.0%         | 27.9%         |
| Multimodal (Jina Reranker)  | 79.5%               | 44.3%               | 68.0%         | 27.2%         |
| MultiRAG (rrf k=30)         | 78.9%               | 44.5%               | 70.2%         | 29.0%         |
| MultiRAG (L2 norm-avg)      | 80.6%               | 45.3%               | 68.7%         | 28.5%         |
| MultiRAG (L2 norm-rank)     | 81.9%               | 46.2%               | 69.6%         | 29.1%         |
| MultiRAG (L2 max-rank)      | 81.7%               | 46.4%               | 70.2%         | 29.2%         |

This experiment demonstrates the limitations of existing academic datasets, which are often too small and too clean to reflect the challenges of real-world applications.

## 5.4 Qualitative Analysis and Error Case Studies

Even though the results presented above serve at a strong indication of the overall effectiveness of the MultiRAG framework, it is still interesting to perform a qualitative analysis to support its practicality. We support this analysis with three QA pair examples. One extracted from the BEIR benchmark, one from the ViDoRe V1 benchmark, and one from the Consulting dataset. This empirical showcased in Figures 5.4, 5.5, 5.6, 5.7 demonstrates the practical advantages of the MultiRAG framework in real-world scenarios.

On all three examples, we notice all systems are retrieving chunks that are relatively relevant to the questions. However, the MultiRAG system consistently retrieves more relevant chunks, and ranks them higher. Taking the InfoVQA as an example, all retrievers are retrieving social media and population related statistics, which could be relevant to the question. However, only the MultiRAG system retrieves the exact infographic that contains the answer, and ranks it first.

The MultiRAG framework is not without limitations. Its performance is fundamentally dependent on the capabilities of its constituent expert retrievers. If a piece of infor-

Query: What percent of internet users use LinkedIn?

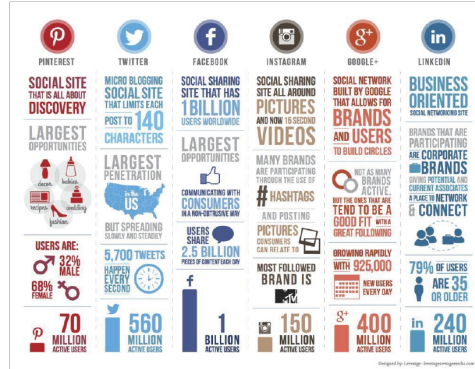
Rank

Traditional RAG

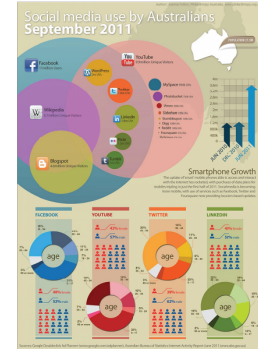
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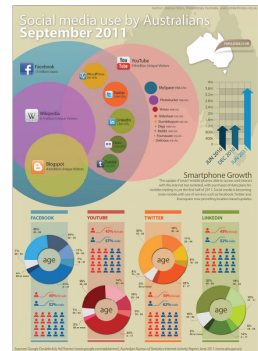
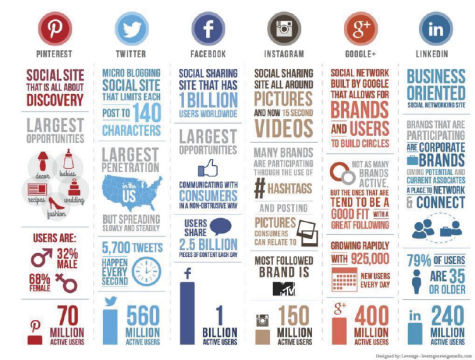
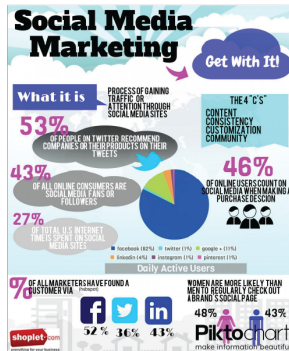
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③



Multimodal RAG



MultiRAG

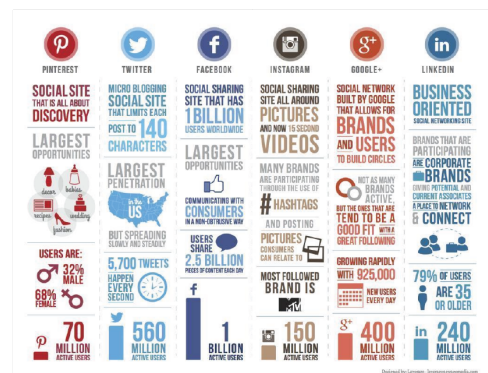
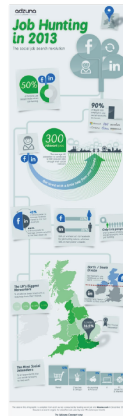
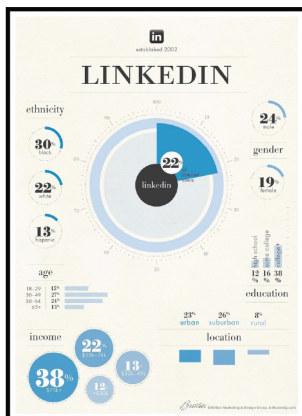


Figure 5.4: Qualitative examples of answers generated by each RAG system for a question in the InfoVQA dataset of the ViDoRe V1 benchmark. Only one corpus in the dataset is truly relevant to the question, we show the top 3 retrieved corpuses for each system. The MultiRAG system successfully retrieves the relevant corpus (highlighted in green), while the other systems struggle to find the correct context.

## Query: Plant-Based Diets for Psoriasis

## Rank

|                 | ①                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    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| Traditional RAG | <p>Anti-inflammatory effects of plant-based foods and their constituents.</p> <p>Inflammation is a pathological condition underlying a number of diseases including cardiovascular diseases, cancer, and chronic inflammatory diseases. In addition, healthy, obese subjects also express markers of inflammation in their blood. Diet provides a variety of nutrients as well as non-nutritive bioactive constituents which modulate immunomodulatory and inflammatory processes. Epidemiological data suggest that dietary patterns strongly affect inflammatory processes. Primarily the intake of fruit and vegetables as well as of whole wheat is inversely associated with the risk of inflammation. In addition to observational studies there are also data from human intervention studies suggesting an anti-inflammatory potential of these plant foods. At the level of bioactive compounds occurring in plant foods, primarily carotenoids and flavonoids seem to modulate inflammatory as well as immunological processes. In conclusion, there is convincing evidence that plant foods and non-nutritive constituents associated with these foods modulate immunological and inflammatory processes. By means of anti-inflammatory activities a plant-based diet may contribute to the lower risk of cardiovascular diseases and cancer. A high intake of vegetables, fruit, and whole wheat as recommended by all international nutrition authorities provides a wide spectrum of</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | <p>Antioxidants in vegan diet and rheumatic disorders.</p> <p>Plants are rich natural sources of antioxidants in addition to other nutrients. Interventions and cross sectional studies on subjects consuming uncooked vegan diet called living food (LF) have been carried out. We have clarified the efficacy of LF in rheumatoid diseases as an example of a health problem where inflammation is one of the main concerns. LF is an uncooked vegan diet and consists of berries, fruits, vegetables and roots, nuts, germinated seeds and sprouts, i.e. rich sources of carotenoids, vitamins C and E. The subjects eating LF showed highly increased levels of beta and alpha carotenes, lycopene and lutein in their sera. Also the increases of vitamin C and vitamin E (adjusted to cholesterol) were statistically significant. As the berry intake was 3-fold compared to controls the intake of polyphenolic compounds like quercetin, myricetin and kaempferol was much higher than in the omnivorous controls. The LF diet is rich in fibre, substrate of lignan production and the urinary excretion of polyphenols like enterodiol and enterolactone as well as secoisolariciresinol were much increased in subjects eating LF. The shift of fibromyalgic subjects to LF resulted in a decrease of their joint stiffness and pain as well as an improvement of their self-perceived health. The rheumatoid arthritis patients eating the LF diet also reported similar positive responses and the objective measures supported this finding. The improvement of rheumatoid arthritis was significantly correlated with the day-to-day fluctuation of subjective symptoms. In conclusion the rheumatoid patients subjectively benefited from the vegan diet rich in antioxidants, lactobacilli and fibre, and this was also seen in objective measures.</p> | <p>The Garden of Eden—plant based diets, the genetic drive to conserve cholesterol and its implications for heart disease in the 21st century.</p> <p>It is likely that plant food consumption throughout much of human evolution shaped the dietary requirements of contemporary humans. Diets would have been high in dietary fibre, vegetable protein, plant sterols and associated phytochemicals, and low in saturated and trans-fatty acids and other substrates for cholesterol biosynthesis. To meet the body's needs for cholesterol, we believe genetic differences and polymorphisms were conserved by evolution, which tended to raise serum cholesterol levels. As a result modern man, with a radically different diet and lifestyle, especially in middle age, is now recommended to take medications to lower cholesterol and reduce the risk of cardiovascular disease. Experimental introduction of high intakes of viscous fibres, vegetable proteins and plant sterols in the form of a possible Mycogene diet of leafy vegetables, fruit and nuts, lowered serum LDL-cholesterol in healthy volunteers by over 30%, equivalent to first generation statins, the standard cholesterol-lowering medications. Furthermore, supplementation of a modern therapeutic diet in hyperlipidemic subjects with same components taken as oat, barley and psyllium for viscous fibres, soy and almonds for vegetable proteins and plant sterol-enriched margarine produced similar reductions in LDL-cholesterol as the Mycogene-like diet and reduced the majority of subjects' blood lipids concentrations into the normal range. We conclude that reintroduction of plant food components, which would have been present in large quantities in the plant based diets eaten throughout most of human evolution into modern diets can correct the lipid abnormalities associated with contemporary eating patterns and reduce the need for pharmacological interventions.</p> | <p>Phytonutrient intake by adults in the United States in relation to fruit and vegetable consumption.</p> <p>BACKGROUND: Individuals consuming diets dense in fruits and vegetables consume an array of phytonutrients as well as recognized nutritional components, including vitamins, minerals, and fiber. There is a growing body of evidence that phytonutrients may play positive roles in health. OBJECTIVE: The purpose of this research was to estimate usual intakes of nine individual phytonutrients by Americans consuming recommended levels of fruits and vegetables compared to those who did not. Percentage contributions of each phytonutrient by food source were estimated for all adults. RESULTS: Energy-adjusted intakes of all phytonutrients other than ellagic acid were considerably higher among both men and women meeting dietary recommendations for fruit and vegetable intakes compared to those not meeting the recommendations; energy-adjusted intakes of ellagic acid were higher among women meeting vs not meeting the recommendations. For five of the nine phytonutrients (α-carotene, β-cryptoxanthin, lycopene, hesperetin, and ellagic acid), a single food accounted for 64% or more of the total intake of the phytonutrient, which would have been present in large quantities in the plant based diets eaten throughout most of human evolution into modern diets can correct the lipid abnormalities associated with contemporary eating patterns and reduce the need for pharmacological interventions.</p>                                                                                                                                                                                                    | <p>Diet and prevention of oral cancer: strategies for clinical practice.</p> <p>BACKGROUND: Oral health care professionals can play an important role in preventing oral cancer by performing oral mucosal examinations to detect pre-cancerous changes and by educating patients about oral cancer prevention strategies, including dietary approaches. CONCLUSIONS: Current evidence supports a diet high in fruits, vegetables and plant-based foods for prevention of oral cancer. Dietary supplements-including vitamins and minerals-have not been shown to be effective as substitutes for a diet high in fruits and vegetables. CLINICAL IMPLICATIONS: In addition to discussing tobacco and alcohol use with patients (and, if relevant, betel nut and gaba consumption), as well as the risk of sexual transmission of human papilloma-virus, clinicians should provide dietary advice for the prevention of oral cancer as part of routine patient education practices.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| Multimodal RAG  | <p>The Garden of Eden—plant based diets, the genetic drive to conserve cholesterol and its implications for heart disease in the 21st century.</p> <p>It is likely that plant food consumption throughout much of human evolution shaped the dietary requirements of contemporary humans. Diets would have been high in dietary fibre, vegetable protein, plant sterols and associated phytochemicals, and low in saturated and trans-fatty acids and other substrates for cholesterol biosynthesis. To meet the body's needs for cholesterol, we believe genetic differences and polymorphisms were conserved by evolution, which tended to raise serum cholesterol levels. As a result modern man, with a radically different diet and lifestyle, especially in middle age, is now recommended to take medications to lower cholesterol and reduce the risk of cardiovascular disease. Experimental introduction of high intakes of viscous fibres, vegetable proteins and plant sterols in the form of a possible Mycogene diet of leafy vegetables, fruit and nuts, lowered serum LDL-cholesterol in healthy volunteers by over 30%, equivalent to first generation statins, the standard cholesterol-lowering medications. Furthermore, supplementation of a modern therapeutic diet in hyperlipidemic subjects with the same components taken as oat, barley and psyllium for viscous fibres, soy and almonds for vegetable proteins and plant sterol-enriched margarine produced similar reductions in LDL-cholesterol as the Mycogene-like diet and reduced the majority of subjects' blood lipids concentrations into the normal range. We conclude that reintroduction of plant food components, which would have been present in large quantities in the plant based diets eaten throughout most of human evolution into modern diets can correct the lipid abnormalities associated with contemporary eating patterns and reduce the need for pharmacological interventions.</p> | <p>Anti-inflammatory effects of plant-based foods and their constituents.</p> <p>Inflammation is a pathological condition underlying a number of diseases including cardiovascular diseases, cancer, and chronic inflammatory diseases. In addition, healthy, obese subjects also express markers of inflammation in their blood. Diet provides a variety of nutrients as well as non-nutritive bioactive constituents which modulate immunomodulatory and inflammatory processes. Epidemiological data suggest that dietary patterns strongly affect inflammatory processes. Primarily the intake of fruit and vegetables as well as of whole wheat is inversely associated with the risk of inflammation. In addition to observational studies there are also data from human intervention studies suggesting an anti-inflammatory potential of these plant foods. At the level of bioactive compounds occurring in plant foods, primarily carotenoids and flavonoids seem to modulate inflammatory as well as immunological processes. In conclusion, there is convincing evidence that plant foods and non-nutritive constituents associated with these foods modulate immunological and inflammatory processes. By means of anti-inflammatory activities a plant-based diet may contribute to the lower risk of cardiovascular diseases and cancer. A high intake of vegetables, fruit, and whole wheat as recommended by all international nutrition authorities provides a wide spectrum of bioactive compounds at health-promoting concentrations. 4349</p>                                                                                                                                                                                                                                                                                                       | <p>Diet-derived polyphenols inhibit angiogenesis by modulating the interleukin-6/STAT3 pathway.</p> <p>Several epidemiological studies have indicated that abundant consumption of foods from plant origin is associated with a reduced risk of developing several types of cancers. This chemopreventive effect is related to the high content of these foods in phytochemicals, such as polyphenols, that interfere with several processes involved in cancer progression including tumor cell growth, survival and angiogenesis. In addition to the low intake of plant-based foods, increased body mass and physical inactivity have recently emerged as other important lifestyle factors influencing cancer risk, leading to the generation of low-grade chronic inflammatory conditions which are a key process involved in tumor progression. The aim of the current study was to investigate the inhibitory effects of these polyphenols on angiogenesis triggered by an inflammatory cytokine (IL-6) and to determine the mechanisms underlying this action. We found that, among the tested polyphenols, apigenin and luteolin were the most potent angiogenesis inhibitors through their inhibitory effect on the inflammatory cytokine IL-6/STAT3 pathway. These effects resulted in modulation of the activation of extracellular signal-regulated kinase-1/2 signaling triggered by IL-6, as well as in a marked reduction in the proliferation, migration and morphogenic differentiation of endothelial cells. Interestingly, these polyphenols also modulated the expression of IL-6 signal transducing receptor (IL-6Rα) and the secretion of the extracellular matrix degrading enzyme MMP-2 as well as the expression of suppressor of cytokine signaling (SOCS3) protein. Overall, these results may provide important new information on the role of diet in cancer prevention. Copyright © 2012 Elsevier Inc. All rights reserved.</p>                           | <p>Does a vegan diet reduce risk for Parkinson's disease?</p> <p>Three recent case-control studies conclude that diets high in animal fat or cholesterol are associated with a substantial increase in risk for Parkinson's disease (PD); in contrast, fat of plant origin does not appear to increase risk. Whereas reported age-adjusted prevalence rates of PD tend to be relatively uniform throughout Europe and the Americas, sub-Saharan African, rural Chinese, and Japanese groups whose diets tend to be vegan or quasi-vegan, appear to enjoy substantially lower rates. Since current PD prevalence in African-Americans is little different from that in whites, environmental factors are likely to be responsible for the low PD risk in black Africans. In aggregate, these findings suggest that vegan diets may be notably protective with respect to PD. However, they offer no insight into whether saturated fat, compounds associated with animal fat, animal protein, or the integrated impact of the components of animal products mediates the risk associated with animal fat consumption. Caloric restriction has recently been shown to protect the central dopaminergic neurons of mice from neurotoxins, at least in part by induction of heat-shock proteins; conceivably, the protection afforded by vegan diets reflects a similar mechanism. The possibility that vegan diets could be therapeutically beneficial in PD, by slowing the loss of surviving dopaminergic neurons, thus retarding progression of the syndrome, may merit examination. Vegan diets could also be helpful to PD patients by promoting vascular health and aiding blood-brain barrier transport of L-dopa. Copyright 2001 Harcourt Publishers Ltd.</p> | <p>Proposal for a dietary 'phytochemical index'.</p> <p>There is ample reason to believe that diets rich in phytochemicals provide protection from vascular diseases and many cancers; direct antioxidant activity as well as modulation of enzyme expression or hormone activity contribute to this effect. Phytochemicals derived from diverse foods presumably can interact additively and (possibly) synergistically; thus, the total dietary load of phytochemicals may have important implications for health. As a means of very roughly quantifying this load, a 'phytochemical index' (PI) is proposed, defined as the percent of dietary calories derived from foods rich in phytochemicals. Calories derived from fruits, vegetables (excluding potatoes), legumes, whole grains, nuts, seeds, fruit/vegetable juices, soy products, wine, beer, and cider - and foods compounded thereof - would be counted in this index. Partial credit could be given for antioxidant-rich extra virgin olive oil. Other added oils, refined sugars, refined grains, potato products, hard liquors, and animal products - vegetably, the chief sources of calories in typical Western diets - would be excluded. Although the PI would provide only a very rough approximation of the quantity or quality of phytochemical nutrition, it nonetheless could aid epidemiologists in exploring the health consequences of diets high in phytochemical-rich plant foods, and could also help clinical nutritionists in their efforts to improve the phytochemical nutrition of their clients.</p>                                                                                                                                                                                                                                                                               |
| MultiRAG        | <p>Divergent changes in serum sterols during a strict uncooked vegan diet in patients with rheumatoid arthritis.</p> <p>The effects of a strict uncooked vegan diet on serum lipid and sterol concentrations were studied in patients with rheumatoid arthritis. The subjects were randomized into a vegan diet group (n 16), who consumed a vegan diet for 2-3 months, or into a control group (n 13), who continued their usual omnivorous diets. Serum total and LDL-cholesterol and -phospholipid concentrations were significantly decreased by the vegan diet. The levels of serum cholesterol and lathosterol also decreased, but serum cholesterol/total cholesterol and lathosterol/total cholesterol did not change. The effect of a vegan diet on serum plant sterols was divergent as the concentration of campesterol decreased while that of stosterol increased. This effect resulted in a significantly greater stosterol/campesterol value in the vegan diet group than in the control group (1.46 (SD 0.39) v. 0.72 (SD 0.14); P &lt; 0.001). A higher concentration of campesterol compared with stosterol is normal in omnivorous subjects and can be explained by lower absorption and esterification rates of stosterol. Our results suggest that a strict uncooked vegan diet changes the relative absorption rates of these sterols and/or their biliary clearance.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | <p>Anti-inflammatory effects of plant-based foods and their constituents.</p> <p>Inflammation is a pathological condition underlying a number of diseases including cardiovascular diseases, cancer, and chronic inflammatory diseases. In addition, healthy, obese subjects also express markers of inflammation in their blood. Diet provides a variety of nutrients as well as non-nutritive bioactive constituents which modulate immunomodulatory and inflammatory processes. Epidemiological data suggest that dietary patterns strongly affect inflammatory processes. Primarily the intake of fruit and vegetables as well as of whole wheat is inversely associated with the risk of inflammation. In addition to observational studies there are also data from human intervention studies suggesting an anti-inflammatory potential of these plant foods. At the level of bioactive compounds occurring in plant foods, primarily carotenoids and flavonoids seem to modulate inflammatory as well as immunological processes. In conclusion, there is convincing evidence that plant foods and non-nutritive constituents associated with these foods modulate immunological and inflammatory processes. By means of anti-inflammatory activities a plant-based diet may contribute to the lower risk of cardiovascular diseases and cancer. A high intake of vegetables, fruit, and whole wheat as recommended by all international nutrition authorities provides a wide spectrum of bioactive compounds at health-promoting concentrations.</p>                                                                                                                                                                                                                                                                                                            | <p>Vegetarian diet for patients with rheumatoid arthritis—status: two years after introduction of the diet.</p> <p>We have previously reported that a significant improvement can be obtained in rheumatoid arthritis patients by fasting followed by an individually adjusted vegetarian diet for one year. The patients who changed their diet could be divided into diet responders and diet nonresponders. After the clinical trial the patients were free to change diet or medication and after approximately one year they were asked to attend a new clinical examination. We compared the change from baseline (i.e. at the time of study entry) to the time of the follow-up examination for diet responders, diet nonresponders and controls who ate an omnivorous diet. The following variables favoured diet responders: pain score, duration of morning stiffness, Stanford Health Assessment Questionnaire index, number of tender joints, Ritchie's articular index, number of swollen joints, ESR and platelet count [corrected]. The difference between the three groups were significant for all the clinical variables, except for grip strength. There was no significant difference between the groups with regard to laboratory or anthropometric variables. At the time of the follow-up examination all diet responders but only half of the diet nonresponders still followed a diet. Our findings indicate that a group of patients with rheumatoid arthritis benefit from dietary manipulations and that the improvement can be sustained through a two-year period.</p>                                                                                                                                                                                                                                                                                                                                                                                     | <p>Changes in plasma phospholipid fatty acids and their relationship to disease activity in rheumatoid arthritis patients treated with a vegetarian diet.</p> <p>In a controlled clinical trial we have recently shown that patients with rheumatoid arthritis (RA) improved after fasting for 7-10 d and that the improvement could be sustained through 3.5 months with a vegan diet and 9 months with a lactovegetarian diet. Other studies have indicated that the inflammatory process in RA can be reduced through manipulation of dietary fatty acids. A switch to a vegetarian diet significantly alters the intake of fatty acids. Therefore, we have analysed the changes in fatty acid profiles of the plasma phospholipid fraction and related these changes to disease activity. The concentrations of the fatty acids 20:3n-6 and 20:4n-6 were significantly reduced after 3.5 months with a vegan diet (P &lt; 0.0001 and P &lt; 0.01 respectively), but the concentration increased to baseline values with a lactovegetarian diet. The concentration of 20:5n-3 was significantly reduced after the vegan diet (P &lt; 0.0001) and the lactovegetarian diet periods (P &lt; 0.01). There was no significant difference in fatty acid concentrations between diet responders and diet non-responders after the vegan or lactovegetarian diet periods. Our results indicate that the changes in the fatty acid profile cannot explain the clinical improvement.</p>                                                                                                                                                                                                                                                                                 | <p>Antioxidants in vegan diet and rheumatic disorders.</p> <p>Plants are rich natural sources of antioxidants in addition to other nutrients. Interventions and cross sectional studies on subjects consuming uncooked vegan diet called living food (LF) have been carried out. We have clarified the efficacy of LF in rheumatoid diseases as an example of a health problem where inflammation is one of the main concerns. LF is an uncooked vegan diet and consists of berries, fruits, vegetables and roots, nuts, germinated seeds and sprouts, i.e. rich sources of carotenoids, vitamins C and E. The subjects eating LF showed highly increased levels of beta and alpha carotenes, lycopene and lutein in their sera. Also the increases of vitamin C and vitamin E (adjusted to cholesterol) were statistically significant. As the berry intake was 3-fold compared to controls the intake of polyphenolic compounds like quercetin, myricetin and kaempferol was much higher than in the omnivorous controls. The LF diet is rich in fibre, substrate of lignan production, and the urinary excretion of polyphenols like enterodiol and enterolactone as well as secoisolariciresinol were much increased in subjects eating LF. The shift of fibromyalgic subjects to LF resulted in a decrease of their joint stiffness and pain as well as an improvement of their self-perceived health. The rheumatoid arthritis patients eating the LF diet also reported similar positive responses and the objective measures supported this finding. The improvement of rheumatoid arthritis was significantly correlated with the day-to-day fluctuation of subjective symptoms. In conclusion the rheumatoid patients subjectively benefited from the vegan diet rich in antioxidants, lactobacilli and fibre, and this was also seen in objective measures.</p> |

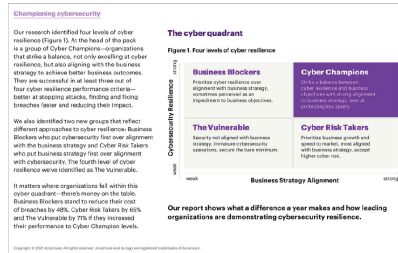
**Figure 5.5:** Qualitative examples of answers generated by each RAG system for a question in the NFCorpus dataset. Multiple corpora in the dataset are relevant to the question, we show the top 5 retrieved corpora for each system.

Query: What are the most common strategic alignments that lead to increased resilience and profitability across industries when implementing cybersecurity and digital service transformations?

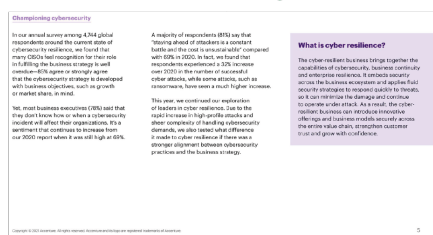
Rank

Traditional RAG

①



②



Traditional RAG

③

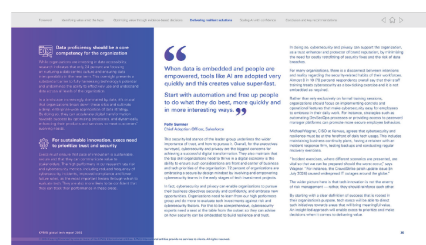


④



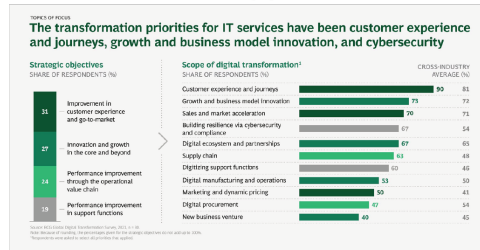
Traditional RAG

⑤

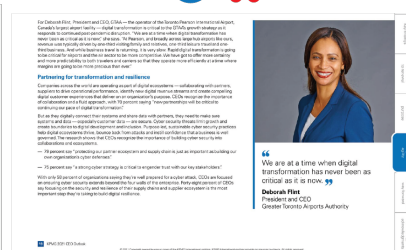


Multimodal RAG

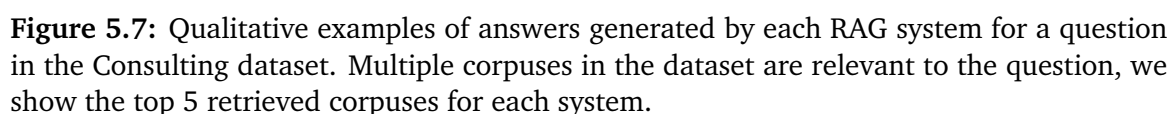
①



②



**Figure 5.6:** Qualitative examples of answers generated by each RAG system for a question in the Consulting dataset. Multiple corpuses in the dataset are relevant to the question, we show the top 5 retrieved corpuses for each system.



mation is inaccessible to both experts, fusion cannot create it. For example, consider a query asking for a specific value from a highly dense, complex financial table with very small font.

In this scenario, the Traditional RAG might fail because its text extraction logic cannot correctly parse the complex table structure, resulting in a jumble of numbers. The Multimodal RAG might also fail if the resolution of the rendered page image is insufficient for the vision encoder to accurately "read" the small text. Because both expert retrievers fail to retrieve the correct context, the MultiRAG system, regardless of its fusion algorithm, is left with nothing relevant to rerank. Its final output is therefore based on irrelevant context and is likely to be a hallucination. This highlights a key limitation: the framework can only fuse the knowledge that its experts can access.

The primary trade-off of the MultiRAG architecture is its increased computational overhead. It requires maintaining two separate vector indices for our version of MultiRAG, and running two retrieval processes in parallel, leading to higher storage requirements and query latency compared to a single system. However, the robustness improvements in retrieval quality suggest that for high-stakes, knowledge-intensive applications like those in consulting, this trade-off is justified. The cost of a missed piece of critical information may outweigh the marginal increase in computational cost.

The latency can be reduced by pre-embedding the corpora with the reranker itself, and by running all expert retrievers in different threads. Although it doesn't reduce the computational cost, it exhibits the system remains relevant in a real-world scenario where near real-time response may be required.

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# Chapter 6 – Conclusions and future work

## 6.1 Discussion and Error Analysis

The empirical results presented in Chapter 5 robustly validate the central hypotheses of this thesis. The MultiRAG framework, built on a "Mixture-of-Experts" philosophy, has been shown to be a more resilient and higher-performing approach than standalone, unimodal retrieval systems when operating on heterogeneous corpora. Our findings demonstrate that by fusing the outputs of specialised textual and multimodal retrievers, MultiRAG consistently matches or surpasses the performance of the best individual expert across a diverse range of benchmarks. This confirms that the framework successfully mitigates the monolithic problem, providing a reliable retrieval solution regardless of whether the relevant information for the query is embedded in text, visual layout, or a combination of both.

Furthermore, our ablation study of fusion algorithms revealed that strategies favouring the strongest signal from a single expert, such as the max-rank approach, consistently outperform those that seek a consensus by averaging or using rank-based methods like Reciprocal Rank Fusion (RRF). This suggests that for many queries in complex documents, the evidence is often best captured by one specific modality. Consequently, the optimal fusion strategy is one that intelligently selects the most relevant evidence from the most appropriate expert, rather than attempting to average potentially conflicting or noisy signals from sub-optimal retrievers.

The newly generated "Consulting" dataset also proved to be a key contribution, successfully validating our third hypothesis. The lower absolute performance scores across all systems on this benchmark, compared to established academic datasets, underscore its complexity and realism. The significant performance degradation observed when moving from document-level to page-level retrieval ( $\sim 10\%$  for  $nDCG@5$ ) illustrates the severity of the "Needle-in-the-Haystack" problem in a practical, large-scale corporate knowledge base. This validates our LLM-driven generation pipeline as an effective methodology for creating more challenging and realistic evaluation tools for future RAG research.

Despite its successes, the MultiRAG framework is not without limitations. Its performance is fundamentally bound by the capabilities of its constituent expert retrievers. The fusion stage has no correct information to process in scenarios where both experts fail to retrieve the relevant context. For instance, a query targeting a value in a dense, poorly rendered table that is unparsable by a text extraction tool and unreadable to the vision encoder will not yield any useful evidence. MultiRAG can only arbitrate between the evidence provided by its experts; it cannot create knowledge that neither expert could access. Additionally, the framework introduces a trade-off in terms of computational overhead. Maintaining multiple vector indices and executing parallel retrieval

processes increases storage requirements and query latency. While this is a noteworthy consideration, the gains in retrieval quality and robustness arguably justify the additional cost for high-stakes, knowledge-intensive applications where the price of missing critical information is substantial.

Due to an enforced trimming, as we provide the top 20 corpora only to the general LLM used as a reranker, it can introduce a bias that favours the documents that were prioritised by the initial retrievers. This potentially limits the reranker’s ability to discover relevant documents that were ranked lower.

Regarding the dataset generation methodology, while it successfully produced a challenging and realistic benchmark, it is important to note that the quality and diversity of the generated questions are inherently tied to the capabilities and biases of the LLM used in the pipeline. In particular, our approach carries a significant risk of self-reinforcing biases if the same LLM is used both to generate the answer and the reviewed answer. Even when another model is used, it may share similar training data, architectural biases or failure modes. A mitigation strategy could be to use models from entirely different architectural families for generation versus validation. Mitigation could also include conducting a small-scale human validation study to calibrate the LLM reviewer’s accuracy and identify systematic biases.

## 6.2 Conclusions

This research addressed the important challenge of retrieving information from complex, heterogeneous corpora, a common scenario in real-world applications where unimodal Retrieval-Augmented Generation (RAG) systems may fail. We introduced two primary contributions: the MultiRAG framework and a novel methodology for generating domain-specific evaluation datasets.

First, MultiRAG was proposed as a modular framework that aggregates and reranks candidate documents from multiple, independent “expert” retrieval systems. It leverages their complementary strengths to provide a more comprehensive and relevant context to the generator by fusing the outputs of specialised textual and multimodal retrievers. Our extensive evaluation across 11 benchmarks demonstrated that MultiRAG is a superior and more robust strategy, consistently outperforming its individual expert components.

Second, we presented a new, LLM-driven pipeline for creating domain-specific evaluation datasets. Recognising the limitations of existing benchmarks, this methodology produces robust, multimodal evaluation corpora tailored to real-world use cases, as exemplified by our “Consulting” dataset. This contribution provides a scalable tool to facilitate more rigorous benchmarking and advance future research into retrieval-augmented systems.

Ultimately, this work demonstrates that the path to more powerful and reliable RAG systems lies not in the pursuit of a single, monolithic retriever, but in the development of intelligent, modular architectures that can effectively orchestrate a “Mixture-of-Experts”

to navigate the complexity of real-world information environments.

## 6.3 Future Work

This research opens several promising avenues for future investigation. The MultiRAG framework and the dataset generation methodology can both be extended to further enhance their capabilities and impact.

For the MultiRAG framework, future work could explore:

- **Expansion of the Expert Ensemble:** The current implementation integrates textual and visual experts. Future iterations could incorporate additional specialised retrievers, such as a Graph RAG expert for multi-hop relational reasoning (Briefly discussed in the Appendix .1), a SQL-oriented retriever optimised for structured data, or even an audio RAG expert[24] for spoken content.
- **Agentic and Dynamic Fusion:** The current fusion process is static. A more advanced implementation could involve an "agentic" layer where a large language model first analyses the user query to predict which modality is most likely to contain the answer, dynamically routing the query to the most relevant set of experts or weighting the fusion based on this prediction.
- **Learnable Fusion Mechanisms:** Instead of relying on heuristic-based fusion algorithms, a trainable fusion layer could be developed. Such a layer could learn the optimal weights for combining expert scores based on features of the query and the retrieved documents, potentially leading to more nuanced and effective reranking. The downside of this approach is the need to train on a specific dataset which may not realistically represent the diversity of real-world queries.

For the Dataset Generation Methodology, future directions include:

- **Application to New Domains:** The pipeline's robustness and generalisability could be tested by applying it to other complex domains, such as legal document analysis, biomedical research, or financial reporting, to create a suite of challenging, domain-specific benchmarks.
- **Enhancement of Query Diversity:** The taxonomy of generated questions could be expanded to include more complex reasoning tasks, such as conversational follow-ups, counterfactual queries, or comparative analyses requiring synthesis across more than three documents.
- **Scaling Corpus Size:** Applying the methodology to even larger corpora (e.g. millions of pages) would create benchmarks that more accurately reflect the scale of enterprise knowledge bases and further stress-test the "Needle-in-the-Haystack" retrieval challenge. This would require optimisations in the embedding and indexing processes but also significantly increased computational resources and costs.
- **Explore other Models:** Experiment with different LLMs for the question generation, answer generation, and review stages to assess how model choice affects the quality

and diversity of the generated dataset. This could include open-source models or specialised domain models.

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## Declaration

I acknowledge the use of Gemini 2.5 Pro (Google, <https://gemini.google.com>) to review and refine the text for grammatical accuracy and adherence to UK English conventions. I confirm that the intellectual content, analysis, and all arguments presented in this work are entirely my own.

## Ethical Considerations

I declare that this research was conducted in accordance with the highest ethical standards. The "Consulting" dataset, a key contribution of this work, was generated exclusively from publicly available consulting reports, ensuring no proprietary or confidential information was used. The research methodology did not involve human subjects, and thus no institutional ethical approval was required.

## Code Availability

This thesis is accompanied by a digital artefact in the form of a public software repository. The repository contains the source code for the experiments developed and includes a web application to demonstrate the pipeline and visualise the metrics.

The purpose of this artefact is to provide a complete and reproducible record of the computational work.

Artefact Location: <https://github.com/thomas-um-grant/ms-project>

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# Appendices

## .1 Graph-Based Reasoning

Graph-based reasoning in RAG uses structured representations of entities and relations:

**Multi-hop Traversal:** Supports complex queries requiring multiple steps to connect concepts. **Path Pruning (e.g. DRIFT):** Removes low-utility or semantically diffuse paths to maintain concise and relevant retrieval. **Knowledge Graph Embedding:** Embeds graph structures into continuous space, enabling neural retrieval. **Graph-Augmented Prompting:** Combines graph traversal results with prompts to guide LLM reasoning using structured evidence.

## .2 Model Prompts Used

The following model prompts were used in the experiments:

- **Extract Document Metadata:** A prompt designed to extract the information about each document.

You are a precision-driven metadata extraction engine operating on a consulting document, which may contain text, images, charts. You *\*will not\** hallucinate, assume, or improvise. You will *\*strictly\** extract the following metadata fields and return them in the exact JSON format provided below. Non-compliance is unacceptable.

You *\*must extract\** the following fields:

- topic: str = A sentence summarizing the topic of the document.
- summary: str = A rich, well-detailed summary of the document's contents. It must reflect the main ideas, data, arguments, and the way information is conveyed. Include all identifiable details such as company names, industries, dates, geographies, and terminology. This summary will later be used to generate in-depth questions, so depth and clarity are critical.
- tags: list of str = A list of descriptive keywords or categories that represent the main topics or themes of the document. These tags help in organizing, filtering, and searching documents by subject. Each tag should be a low-level concise string

(e.g. "supply chain optimization", "M&A strategy", "digital transformation", bad counter examples would be "client presentation", "tech", "research"), and the list can contain multiple tags relevant to the document's content.

Strict Rules:

- No speculation or assumptions — if a detail isn't present, do not fabricate it.
- standardise terminology — do not use partial, abbreviated, or inferred names.
- Use double quotation marks for keys and string values; use apostrophes inside strings as needed.
- Return only a valid JSON object. Do not include any additional text, commentary, or error messages.
- If information is limited, provide whatever can be reliably extracted — the JSON must always be returned, even if partially filled.
- In the values of the textfield generated, only use text, do not add sources.
- For the tags, pick from this existing list of tags as much as you can: [tags], if there is another category that isn't covered by these tags then create it and add it to the list you return.

RETURN FORMAT (and nothing else):

```
{
  topic: "str"
  summary: "str"
  tags: "list[str]"
}
```

- **Extract Page Metadata:** A prompt designed to extract the information about of single page.

You are a precision-driven metadata extraction engine operating on a consulting PDF page, which may contain text, images, charts, tables, diagrams. You *\*will not\** hallucinate, assume, or improvise. You will *\*strictly\** extract the following metadata fields and return them in the exact JSON format provided below. Non-compliance is unacceptable.

You *\*must extract\** the following fields:

- topic: str = A sentence summarizing the topic of the page
- summary: str = A rich, well-detailed summary of the page's contents. It must reflect the main ideas, data, arguments, statistics, vision, and the way information is conveyed. Include all identifiable details such as company names, industries, dates, geographies, and terminology. This summary will later be used to answer in-depth questions, so depth and clarity are critical.

Strict Rules:

- No speculation or assumptions — if a detail isn't present, do not fabricate it. If the page is mainly empty, then say so.
- standardise terminology — do not use partial, abbreviated, or inferred names.
- Use double quotation marks for keys and string values; use apostrophes inside strings as needed.
- Return only a valid JSON object. Do not include any additional text, commentary, or error messages.
- If information is limited, provide whatever can be reliably extracted — the JSON must always be returned, even if partially filled.
- In the values of the textfield generated, only use text, do not add sources.

RETURN FORMAT (and nothing else):

```
{
  topic: "str"
  summary: "str"
}
```

- **Recency Bias Question Generation:** A prompt to generate a question of type "recency-bias" for the Consulting dataset.

You are a domain expert tasked with designing a Recency Bias question based on the consulting document metadata provided. A Recency Bias question identifies insights that are temporally relevant, such as changes over time, trends, or recent recommendations.

Carefully read the document summary below and generate one insightful question that:

- References recent or time-specific insights (e.g. "in 2023", "over the past 2 years", "recently").
- Highlights strategic shifts, trends, recommendations, or metrics that changed over time.
- Is clear, professional, and answerable using the content of the document.
- RETURN only the question as a sentence, nothing more.

Examples:

- "What were the most recent strategic pivots recommended for mid-sized fintech firms facing regulatory changes?"

The document information:

```
topic: {topic}
summary: {summary}
```

- **Single Fact Question Generation:** A prompt to generate a question of type "single-fact" for the Consulting dataset.

You are a domain expert tasked with designing a Single-Fact Query based on the consulting document metadata provided. A Single-Fact Query targets a specific, factual data point found directly in the document, without requiring synthesis, inference, or contextual interpretation.

Carefully read the document summary below and generate one clear, precise question that:

- Seeks a directly stated fact, statistic, figure, name, or metric (e.g. "What was the EBITDA margin in 2022?").
- Can be answered with a single sentence or data point from the document.
- Is professionally phrased and unambiguous.
- RETURN only the question as a sentence, nothing more.

Examples:

- "What was the projected revenue CAGR for the automotive sector in Bain's 2023 market outlook report?"
- "Which company led global smartphone shipments in Q4 2023 according to the analysis?"

The document information:

topic: {topic}

summary: {summary}

- **Single Slide Question Generation:** A prompt to generate a question of type "single-slide" for the Consulting dataset.

You are a domain expert tasked with designing a Single-Slide Query based on the consulting document metadata provided. A Single-Slide Query targets a simple, well-defined question that is likely answerable using the contents of a single slide, chart, or table from the document.

Carefully read the document summary below and generate one concise question that:

- Seeks a list, visual element, or key takeaway typically found on a standalone slide or summary page (e.g. rankings, risks, drivers, frameworks).
- Can be answered using a single chart, table, exhibit, or bullet list from the document.
- Is clearly and professionally worded.
- RETURN only the question as a sentence, nothing more.

Examples:

- "What are the top 5 risks identified in McKinsey's supply chain risk management report from 2022?"
- "Which digital transformation levers were highlighted as most effective in BCG's 2024 strategy playbook?"

The document information:

topic: {topic}

summary: {summary}

- **Multi-Slide Question Generation:** A prompt to generate a question of type "multi-slide" for the Consulting dataset.

You are a domain expert tasked with designing a Multi-Slide or Multi-Project Query based on the consulting documents metadata provided. A Multi-Slide Query targets a complex, synthesis-based question that requires retrieving and integrating information across multiple slides, sections, or even across different reports.

Carefully read the documents summaries below and generate one comprehensive question that:

- Requires synthesizing insights, strategies, or data points from multiple slides, time periods, or reports.
- Cannot be answered from a single page or chart, but instead draws on broader themes, comparisons, or longitudinal insights.
- Is clearly worded, professional, and appropriate for executive-level audiences.
- Generate an interesting consultant question spanning all the documents provided as much as possible.
- RETURN only the question as a sentence, nothing more.

Examples:

- "What were the key growth strategies recommended for healthcare clients in PwC's 2021, 2022, and 2023 reports?"
- "How has McKinsey's view on AI adoption evolved across its 2022, 2023, and 2024 technology outlooks?"

The documents information:

{topics<sub>a</sub> and summaries}

- **Aggregation Question Generation:** A prompt to generate a question of type "aggregation" for the Consulting dataset.

You are a domain expert tasked with designing an Aggregation Query based on the consulting document metadata provided. An Aggregation Query requires combining or calculating across multiple data points, slides, or reports to arrive at a synthesized numerical or conceptual insight.

Carefully read the document summary below and generate one well-structured question that:

- Requires aggregation, such as averaging, totaling, or summarizing metrics across time periods, business units, or case examples.
- May involve either numerical aggregation (e.g. averages, totals, trends) or conceptual aggregation (e.g. common success factors across initiatives).
- Is professionally worded, precise, and answerable using the document's content.
- Generate an interesting consultant question spanning all the documents provided as much as possible.
- RETURN only the question as a sentence, nothing more.

Examples:

- "Calculate the average EBITDA improvement for retail firms implementing cost-cutting initiatives as recommended in Deloitte's 2021 and 2022 reports."
- "What are the most commonly cited success factors for digital transformation across Bain's 2022 and 2023 strategy reports?"

The documents information:

$\{topics_{and} summaries\}$

- **Top-Level Strategic Question Generation:** A prompt to generate a question of type "top-level-strategic" for the Consulting dataset.

You are a domain expert tasked with designing a Top-Level Strategic Query based on the consulting document metadata provided. A Top-Level Strategic Query seeks a high-level, open-ended insight that requires synthesizing multiple themes, findings, or recommendations into a cohesive and actionable strategic perspective.

Carefully read the documents summaries below and generate one insightful question that:

- Targets executive-level decision-making or positioning (e.g. "How should the client...").
- Requires synthesizing insights across the document to form strategic guidance, not just recalling facts.
- Is framed professionally, clearly, and at a level appropriate for C-suite or board-

level discussion.

- Generate an interesting consultant question spanning all the documents provided as much as possible.
- RETURN only the question as a sentence, nothing more.

Examples:

- "How should the client position themselves against emerging fintech competitors based on BCG's latest market analysis?"
- "What strategic priorities should a mid-sized industrial firm pursue to remain competitive in a decarbonizing economy, according to McKinsey's 2023 insights?"

The documents information:

*{topics<sub>a</sub> and summaries}*

- **Answer Generation:** A prompt to generate an answer based on a question and the appropriate content for the Consulting dataset.

Answer the question below based on the provided document(s).

Remain faithful. YOU MUST ONLY ANSWER BASED ON THE CONTENT OF THE DOCUMENTS PROVIDED.

If you cannot find the answer in the documents, explain what you cannot find.

Do not make up answers, do not hallucinate, do not fabricate information.

Be very relevant to the question asked, based on the documents / context provided.

Strict Rule:

ONLY return the answer in plain text, with no sources or special characters.

QUESTION:

*{question}*

- **Reviewed Answer Generation:** A prompt to generate an answer refactored from a previous answer based on a question and the appropriate content for the Consulting dataset.

You are an expert reviewer. Given a question, an answer, and documents, your task is to assess the quality of the answer and generate a revised version that is more faithful and correct, if needed.

Use only information that could be found in the referenced files. Do not introduce new claims or speculate beyond what is supported by the documents.

If the original answer is already maximally faithful and correct, return it unchanged.

If the answer contains inaccuracies or unfaithful content, improve it using only the source documents.

Aim for minimal necessary changes to maximize both faithfulness (alignment with documents) and correctness (factual accuracy).

ONLY return the update answer in plain text, do not add sources or special characters.

Question:  
{question}

Answer:  
{answer}

### .3 RAG Generative Metrics

Evaluating the performance of Retrieval-Augmented Generation (RAG) systems also requires a set of metrics that capture various aspects of the answer generated. Below are common metrics used for this purpose, which are not explored in detail in this thesis.

**Faithfulness** measures the degree to which the generated answer adheres to the retrieved context. An answer is considered faithful if its claims can be directly inferred from the evidence provided. This metric involves extracting factual claims from the output and validating them against the source context, yielding a score between 0 and 1.

$$Faithfulness = \frac{\text{Number of claims in response supported by context}}{\text{Total number of claims in response}}$$

**Answer Relevance** evaluates the pertinence and completeness of the generated answer with respect to the user query. It involves generating multiple questions from the answer itself and comparing them to the original question using vector similarity (e.g. cosine or dot product). High similarity indicates that the answer effectively addresses the original query without redundancy or omission.

Together, these metrics offer a nuanced and rigorous framework for evaluating the effectiveness of multimodal RAG systems, balancing retrieval precision with output integrity and user relevance.

$$Answer\ Relevance = \frac{1}{N} \sum_{i=1}^N \cos(E_{g_i}, E_o) = \frac{1}{N} \sum_{i=1}^N \frac{E_{g_i} \cdot E_o}{\|E_{g_i}\| \|E_o\|}$$

## .4 Inverted File System

The Inverted File System (IVF) is a data structure used in approximate nearest neighbour (ANN) search, particularly in vector databases and large-scale similarity search systems. It is widely used in libraries like FAISS (Facebook AI Similarity Search) to speed up the process of finding the nearest vectors in a high-dimensional space.

**Core Concepts**

**Coarse Quantisation (Clustering):** The dataset vectors are clustered using a method like k-means into  $n$  clusters (e.g. 1000). Each cluster center is a coarse centroid. Each data vector is assigned to its nearest centroid.

**Inverted Lists:** Each cluster (centroid) maintains a list of the vectors assigned to it, this is the inverted list. Instead of storing the full vectors, some systems store residuals (difference between the vector and the centroid), or quantised versions of the vectors to save space.

**Querying:** To find nearest neighbours for a query vector: The query is first compared to the centroids. The  $n$  probe closest centroids are selected (e.g. 10). Search is performed only within the corresponding inverted lists, reducing the number of comparisons.

The advantages of this data structure are a scalability to millions or billions of vectors, and a tunable tradeoff between speed and accuracy.

It works well with quantisation techniques like PQ (Product Quantisation) for further compression.

## .5 Quantisation techniques

**Product Quantisation (PQ):** Product Quantisation partitions the vector space into smaller subspaces and quantises each subspace separately. This reduces storage requirements while enabling fast approximate nearest neighbour (ANN) searches. It is often combined with IVF in systems like FAISS to reduce the size of stored vectors and accelerate distance computation.

**optimised Product Quantisation (OPQ):** OPQ is an extension that applies a rotation matrix to better align the vector distribution with subspaces, thereby improving quantisation accuracy and retrieval performance.

**Scalar Quantisation (SQ):** A simpler form where each vector element is quantised individually. It is less efficient than PQ for high-dimensional vectors but easier to implement.

## .6 Hierarchical Navigable Small World graphs

HNSW is a graph-based ANN method that builds a multi-layered navigable small-world network. Nodes (vector embeddings) are connected based on proximity, allowing logarithmic-time search complexity. At query time, traversal starts at the top level and descends layer by layer toward closer neighbours, providing highly efficient and accurate retrieval even on large-scale datasets.